

Representation Multiplicity in Causal Forests

Yi Niu*

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Abstract

Feature-subsampled causal forests draw random sets of covariate columns, even when several columns encode the same split action. Equivalent recodings can therefore alter candidate-split probabilities without changing covariate information. I characterize the forest target by the cumulative resolution intensity of an unresolved effect component along a tree path. In a randomized two-modifier design, growing blocks of strictly increasing recodings make the scalar `causal_forest` path in `grf` 2.4.0 converge to representation-dependent targets whose signs disagree for half the population. Every fixed finite representation covered by the theorem recovers the CATE under the stated conditions. Simulations and a cash-transfer ranking audit show that recoding sensitivity can greatly exceed seed sensitivity. A class-sampled forest restores exact invariance on a declared finite array under canonical routing and common class-level randomness.

Keywords: causal forests; feature subsampling; policy learning; randomized search; representation invariance; redundant covariates.

JEL Classification: C14, C21, C55

*Department of Economics, University of Pennsylvania. Email: n1y1@sas.upenn.edu

1 Introduction

Suppose two analysts fit the same causal forest to the same data. The first includes household size. The second retains household size together with its logarithm and several split-equivalent recodings. The two representations contain the same information and induce the same threshold partitions, but feature subsampling treats their columns as separate candidates. Household size therefore receives several chances to enter the split search. Additional trees only estimate the resulting representation-specific infinite-forest limit more precisely.

This distinction matters when forests are used to rank treatment effects. A modifier affects a leaf prediction after the tree selects a split that separates its relevant states. Candidate inclusion alone does not suffice, because a stronger unresolved modifier may win the split. The probability of selection accumulates over the nodes reached before the leaf stops. I call the sum of these history-conditional selection probabilities the component’s cumulative resolution intensity. Its exact resolution probability is one minus the expected survival product. When every one-node probability vanishes, a limiting intensity λ gives resolution probability $1 - e^{-\lambda}$.

The main result, Theorem 2, studies the scalar `causal_forest` path in `grf` 2.4.0 under a two-modifier randomized design. One representation repeats the weaker modifier and leaves the stronger modifier as a singleton. The other reverses their roles. Both representations generate the same sigma-field at every sample size. When the number of recodings grows fast enough relative to path depth, the singleton is unlikely to be selected before stopping. The two forests then converge to different component targets, and their signs disagree for half the population. With fixed `mtry`, the boundary is proportional to path depth. Under the package-default `mtry` scaling, within the same asymptotic learner sequence, it is proportional to squared path depth. Every fixed finite representation eventually lies on the recovery side of the boundary. The inconsistent limits therefore concern growing representation sequences. The same mechanism applies to any fixed number of signed effect modifiers.

The proof uses the general target formula in Theorem 1. For an honest score forest, a component with limiting cumulative intensity λ enters the target with coefficient $1 - e^{-\lambda}$. This coefficient measures the attenuation caused by incomplete access to the component. Applying the formula requires learner-specific leaf, aggregation, and hazard arguments. Section 3 provides those arguments for the signed `grf` design.

The theorem permits exact copies and deterministic strictly increasing recodings, which preserve the source split library and routing rule. The simulations and Kenya exercise ask how large the resulting sensitivity can be in finite samples. In the highest-dimensional Monte Carlo design, split-equivalent recodings produce opposite-signed CATE estimates for

46.2 percent of evaluation observations, compared with 0.09 percent after changing only the forest seed. In public data from the Kenya cash-transfer experiment of Egger et al. [2022], author-constructed preprocessing-style bundles change selected-set membership for 5.9 to 7.5 percent of households under a hypothetical top-half ranking. Both exercises compare the stability of fitted rankings across representations. Assessing accuracy or welfare would require a different design. The recodings were constructed for this paper and did not appear in the original study.

The paper also develops a class-sampled forest for a declared finite array. It certifies columns that generate the same threshold actions on a declared array and samples the resulting classes instead of raw columns. Under class-indexed randomness and canonical routing, Proposition 2 gives exact equality of partitions, predictions, reported standard errors, and selected sets on that array. Extending this equality to new covariate values requires an additional routing rule.

Earlier work documents the underlying candidate-set effect. Duplicate columns enter randomized candidate sets more often and can change prediction or variable importance [Louppe, 2014, Strobl et al., 2007, Tološi and Lengauer, 2011, Gregorutti et al., 2017, Kubuś, 2018]. Existing remedies reduce the influence of redundant features. Chen and Zhang [2013] sample modules before choosing representative variables, whereas Colot et al. [2021] weight features inversely by estimated cluster size. These studies establish the duplicate-column diagnosis and motivate group-aware feature sampling. The question here is how the same mechanism changes causal-forest targets.

High-dimensional forest theory allows the ambient dimension, candidate budget, and path depth to vary. Its recovery bounds depend on candidate access along a path [Chi et al., 2022, Klusowski and Tian, 2024, Mei et al., 2026a]. The index $\mathcal{I}_n = H_n \mathbb{E} Q_{M_n} / M_n$ is the corresponding feature-access measure for the source law studied here, where H_n is path depth, Q_{M_n} is the nodewise candidate count, and M_n is the number of raw columns. The paired representations in Theorem 2 generate the same sigma-field but assign different access probabilities to the same split actions, and the theorem traces this difference into representation-specific causal-forest targets, sign reversals, regret, and a sharp multiplicity boundary for `grf`. Theorem 1 maps resolution hazards into target coefficients, and Theorem 2 verifies its conditions for the source learner.

Proposition 2 sharpens the established grouping remedy. It uses exact split-action certificates and a coupling that preserves every listed forest output on a declared finite array. The result differs from approximate correlation clusters and from exhaustive-CART invariance to monotone recoding [Hong and Choi, 2026]. Related work on forest randomization, honest inference, pointwise convergence, and accuracy provides further background [Biau, 2012,

Klusowski, 2021, Mei et al., 2026b, Wager and Athey, 2018, Athey et al., 2019, Cattaneo et al., 2022, 2025]. The treatment-ranking interpretation follows Hirano and Porter [2009], Kitagawa and Tetenov [2018], and Athey and Wager [2021].

Section 2 develops the hazard formula. Section 3 maps it to the signed `grf` design and derives the multiplicity boundary. Section 4 presents the class-sampled correction, and Section 5 reports the simulation and cash-transfer diagnostics. Appendix A contains the proofs and source verification. The replication archive documents every simulation and empirical recoding.

2 Resolution Hazards and Forest Targets

Let $\{(Y_i, W_i, X_i)\}_{i=1}^n$ be an i.i.d. training sample, where Y_i is the observed outcome, $W_i \in \{0, 1\}$ is treatment, and X_i contains the covariates, and let $\mathcal{D}_n$ denote this sample. Let $Y_i(w)$ denote the potential outcome under treatment w , and let P_X denote the marginal law of X_i . Define

$$\tau(x) = \mathbb{E}\{Y_i(1) - Y_i(0) \mid X_i = x\}.$$

At each internal node, a tree draws candidate covariate columns, evaluates their admissible threshold splits, and selects one using the stated criterion and tie rule. A forest averages predictions across randomized trees. We call this recursive procedure *tree search*.

Let $\mathcal{X}$ denote the support of X . A raw representation is a measurable map from $\mathcal{X}$ to $\mathbb{R}^M$. We write it as $R = (R_1, \dots, R_M)$, with $R(x) = (R_1(x), \dots, R_M(x))$. Throughout the asymptotic results, $R = R_n$ and $M = M_n$ may vary with n . The index is suppressed when it does not aid the exposition. Two representations are information equivalent if

$$\sigma\{R(X)\} = \sigma\{\tilde{R}(X)\}.$$

Tree search requires a more restrictive equivalence. At a node containing index set I , column j generates the unordered threshold-split library

$$\mathcal{H}_j(I) = \{\{L, I \setminus L\} : L = \{i \in I : R_j(X_i) \leq a\}, a \in \mathbb{R}\},$$

under declared tie and missing-value rules. Columns with the same library at every reachable node are *split equivalent*. The resulting equivalence classes are the split-action classes, and G denotes their number.

Three notions of equivalence will be useful. Information equivalence compares sigma-fields. Split equivalence compares the actions available to tree search. Finite-array certifi-

cation, introduced in Section 4, verifies split equivalence only on a declared array. Exact copies satisfy all three notions on every array, but the converses need not hold.

The signed design in Section 3 provides the concrete mechanism. One of two effect components receives c_n raw labels and the other remains a singleton. If the singleton enters the candidate set at a node with probability ω_n and a target path has order H_n internal nodes, then $\mathcal{I}_n = H_n\omega_n$ is its cumulative-inclusion index. Vanishing $\mathcal{I}_n$ leads to omission, while diverging $\mathcal{I}_n$ leads to recovery. The abstract result below expresses the same idea through history-conditional successful-resolution hazards.

To state the abstract transfer result, consider a scalar honest-score forest. Each tree uses a splitting sample to construct an adaptive partition from random subsets of representation columns and an independent estimation sample to compute its leaf moment. Let Θ denote the randomization of a generic tree. The observation, covariate, tree-state, and randomization spaces are standard Borel. All split, tie, stopping, leaf, score, and prediction maps are jointly measurable in their displayed arguments. Under representation R , let I_n^{est} be its estimation indices and let $L_n^R(x)$ be the terminal cell containing x . Let $\mathcal{Z}_n$ be the sigma-field generated by a standard-Borel random element containing the score-training folds, fitted nuisance functions, and their training randomization. Let $\Gamma_n(y, w, x)$ be a scalar score map whose randomness is $\mathcal{Z}_n$ -measurable, and write $\Gamma_{i,n} = \Gamma_n(Y_i, W_i, X_i)$, with the fold for observation i understood when scores are cross-fitted. Let (Y', W', X') be a fresh observation, independent of the training sample and all nuisance and tree randomization. Define

$$N_n^{R,\text{est}}(x) = \sum_{i \in I_n^{\text{est}}} \mathbf{1}\{X_i \in L_n^R(x)\},$$

$$\tilde{T}_n^R(x) = \frac{1}{N_n^{R,\text{est}}(x)} \sum_{i \in I_n^{\text{est}}} \mathbf{1}\{X_i \in L_n^R(x)\} \Gamma_{i,n}.$$

Set $\tilde{T}_n^R(x) = 0$ on an empty estimation leaf. Here $N_n^{R,\text{est}}(x)$ is the number of estimation observations in the terminal cell containing x , and $\tilde{T}_n^R(x)$ is their average score. Empty estimation leaves are pruned, so the denominator is positive whenever a tree contributes at x . Conditional means on population-null leaves may be assigned any fixed version because the statements below are integrated with respect to P_X . For $b = 1, \dots, B_n$, append the index b to these objects when they are generated by tree b . Let $\hat{T}_n^R : \mathcal{X} \rightarrow \mathbb{R}$ be the prediction returned by the B_n -tree forest.

Assumption 1. *Along the representation sequence R_n , there exist a fixed finite number G , a constant μ , and measurable functions $f_1, \dots, f_G : \mathcal{X} \rightarrow \mathbb{R}$ satisfying $\mathbb{E}f_g(X) = 0$ for*

$g = 1, \dots, G$ and $\sum_{g=1}^G \|f_g\|_\infty < \infty$, and a function $T^* : \mathcal{X} \rightarrow \mathbb{R}$ such that

$$T^*(x) = \mu + \sum_{g=1}^G f_g(x),$$

$$\|\mathbb{E}\{\Gamma_n(Y', W', X') \mid X' = \cdot, \mathcal{Z}_n\} - T^*\|_\infty = o_p(1).$$

Here T^* is the limiting conditional score mean. Centering the components sets $\mu = \mathbb{E}T^*(X)$, and f_g records the deviation associated with split-action class g . The decomposition is tied to the declared classes and cannot be recovered from T^* without that structure. The uniform approximation requires the conditional score mean to approach T^* over $\mathcal{X}$, while the bound on $\sum_g \|f_g\|_\infty$ keeps the component sum uniformly bounded. In the signed design below, $T^* = \tau_a$ and the two components are $a_1 s_1$ and $a_2 s_2$. More generally, a component may be an effect modifier, a regression component, a conditional-moment component, or another feature of a scalar generalized-forest score.

For a target x and split-action class g , a split *resolves* the class when it uses that class's threshold action, making the target's descendant leaf locally constant in the corresponding component. Candidate availability alone is not resolution. A failure history records the entire unresolved path up to arrival at the current node. At any nonterminal history, $\rho_{g,n,t}^{R,0}(x)$ is the conditional probability that the next tree transition resolves g . Let $K_{g,n,t}^{R,0}$ be the regular conditional law of the next history given failure to resolve g . Complete-history conditioning imposes no Markov restriction. Standard Borelness gives jointly measurable versions of the finite collection of kernels. If the current hazard is one, that node remains in the product with zero failure factor and the next state is absorbing. Histories reachable only after an earlier null failure are assigned zero hazard and the absorbing transition. Appendix A records the construction.

The potential no-resolution path starts from the original root law and evolves according to $K_{g,n,t}^{R,0}$. On every positive-probability branch it retains the raw dimension, candidate law, stopping rule, and competing actions of the original tree. Let $\mathcal{F}_{g,n,t-1}^{R,0}(x)$ contain the complete failure history upon arrival at node t . The event that node t is reached and its current state are $\mathcal{F}_{g,n,t-1}^{R,0}(x)$ -measurable, making the hazard predictable. Each split sends the target to a strict subset of the finite splitting sample, and a null failure branch is absorbing. The path therefore terminates after finitely many internal nodes. Denote this number by $J_{g,n}^{R,0}(x)$.

On the actual target path, let $S_{g,n,t}^R(x)$ indicate that the split at internal node t resolves class g , and set it to zero after the path stops. The success rule is part of the forest specification. Define

$$D_{g,n}^R(x) = \mathbf{1}\{S_{g,n,t}^R(x) = 1 \text{ for some } t \geq 1\}.$$

A class can enter the candidate set without being selected for the split. Sequential conditioning gives the exact failure-path identity

$$\mathbb{E}_\Theta\{1 - D_{g,n}^R(x) \mid \mathcal{D}_n, \mathcal{Z}_n\} = \mathbb{E}_{\Theta^0} \left[\prod_{t=1}^{J_{g,n}^{R,0}(x)} \{1 - \rho_{g,n,t}^{R,0}(x)\} \mid \mathcal{D}_n, \mathcal{Z}_n \right], \quad (1)$$

for P_X -almost every x , where Θ^0 denotes potential failure-path randomization. The product equals one when $J_{g,n}^{R,0}(x) = 0$. Define the cumulative intensity and maximal one-node hazard by

$$\Lambda_{g,n}^R(x) = \sum_{t=1}^{J_{g,n}^{R,0}(x)} \rho_{g,n,t}^{R,0}(x), \quad \bar{\rho}_{g,n}^R(x) = \max_{t \leq J_{g,n}^{R,0}(x)} \rho_{g,n,t}^{R,0}(x),$$

with both quantities zero when $J_{g,n}^{R,0}(x) = 0$. Thus $\Lambda_{g,n}^R$ is a sum of conditional success probabilities, while the exact conditional failure probability is the expected product in (1).

Expectations bearing the subscript Θ or Θ^0 integrate over actual-tree or potential failure-path randomization, and $\mathbb{P}_X$ denotes probability for an independent evaluation covariate with law P_X . All integrals and $L^p(P_X)$ norms below use the same evaluation law. For a measurable limiting function $\lambda_g^R : \mathcal{X} \rightarrow [0, \infty]$, use the convention $e^{-\infty} = 0$. Convergence to an extended-real limit is understood after the transformation $u \mapsto u/(1+u)$.

Assumption 2. *Along the sequence $R = R_n$, the following conditions hold uniformly over the fixed finite set of split-action classes.*

- (i) *Honest leaf and component reduction. The honest leaf statistic estimates its score mean.*

$$\mathbb{E}_\Theta \left[\int \left| \tilde{T}_n^R(x) - \mathbb{E}\{\Gamma_n(Y', W', X') \mid X' \in L_n^R(x), \mathcal{D}_n, \Theta, \mathcal{Z}_n\} \right| dP_X(x) \mid \mathcal{D}_n, \mathcal{Z}_n \right] = o_p(1). \quad (2)$$

For each component, the conditional leaf mean approaches $f_g(x)$ after resolution and zero before resolution, in the following integrated sense.

$$\sum_{g=1}^G \mathbb{E}_\Theta \left[\int \left| \mathbb{E}\{f_g(X') \mid X' \in L_n^R(x), \mathcal{D}_n, \Theta, \mathcal{Z}_n\} - D_{g,n}^R(x) f_g(x) \right| dP_X(x) \mid \mathcal{D}_n, \mathcal{Z}_n \right] = o_p(1). \quad (3)$$

(ii) Forest aggregation. *The reported forest admits the honest linearization*

$$\left\| \widehat{T}_n^R - \frac{1}{B_n} \sum_{b=1}^{B_n} \widetilde{T}_{b,n}^R \right\|_{L^1(P_X)} = o_p(1), \quad B_n \rightarrow \infty, \quad (4)$$

and, conditional on $(\mathcal{D}_n, \mathcal{Z}_n)$, every tree has the same marginal law as the generic tree Θ . Under this same conditioning, the tree randomizers are independent or form mutually independent groups of uniformly bounded size. For some constant $C < \infty$,

$$\sup_{n,b} \mathbb{E}_\Theta \left[\|\widetilde{T}_{b,n}^R\|_{L^2(P_X)}^2 \middle| \mathcal{D}_n, \mathcal{Z}_n \right] \leq C \quad \text{with probability tending to one.}$$

(iii) Hazard convergence. *Under the joint law of $(\mathcal{D}_n, \mathcal{Z}_n)$, an independent $X \sim P_X$, and a fresh potential failure path generated by the preceding kernels,*

$$\Lambda_{g,n}^R(X) \rightarrow_p \lambda_g^R(X), \quad \bar{\rho}_{g,n}^R(X) \rightarrow_p 0 \quad \text{on } \{\lambda_g^R(X) < \infty\}, \quad (5)$$

Assumption 2 gives the estimator conditions used in Theorem 1. Part (i) links honest leaves to the conditional score mean and records whether each component has been resolved. Part (ii) approximates the reported forest by the average of its linearized trees. Part (iii) requires convergence of the cumulative resolution hazard and excludes a nonnegligible one-node hazard when that limit is finite. Section 3 verifies these conditions for the scalar **grf** path and design stated there.

Assumptions 1 and 2(i) imply the component leaf reduction

$$\mathbb{E}_\Theta \left[\left\| \widetilde{T}_n^R - \mu - \sum_{g=1}^G D_{g,n}^R f_g \right\|_{L^1(P_X)} \middle| \mathcal{D}_n, \mathcal{Z}_n \right] = o_p(1). \quad (6)$$

The next theorem isolates the probabilistic step that is common across learners. Section 3 verifies its assumptions for the scalar **grf** learner and obtains the source-specific result in Theorem 2.

Theorem 1 (Resolution hazards and forest targets). *Under Assumptions 1 and 2,*

$$\left\| \widehat{T}_n^R - \left[\mu + \sum_{g=1}^G \{1 - e^{-\lambda_g^R(\cdot)}\} f_g \right] \right\|_{L^1(P_X)} \rightarrow_p 0. \quad (7)$$

The coefficient function on f_g is $1 - e^{-\lambda_g^R(\cdot)}$. It vanishes with the hazard, lies strictly

between zero and one for a finite positive hazard, and approaches one as the hazard diverges. Information-equivalent representations share a target when their componentwise hazards share a limit. Different hazard limits can produce different targets.

3 A Signed grf Design

3.1 Design and representations

For the signed forest result, let $X = (X_1, X_2) \sim \text{Unif}([0, 1]^2)$, set

$$\iota_j(X) = \mathbf{1}\{X_j \geq 1/2\}, \quad s_j(X) = 2\iota_j(X) - 1,$$

and fix constants $0 < \underline{a} < \bar{a} < \infty$ and $0 < \Delta_a < \bar{a} - \underline{a}$. The coefficient region is

$$\mathcal{C}_a = \{(a_1, a_2) : \underline{a} \leq a_1 < a_2 \leq \bar{a}, a_2 - a_1 \geq \Delta_a\},$$

and

$$\tau_a(X) = a_1 s_1(X) + a_2 s_2(X).$$

For each n and $a \in \mathcal{C}_a$, let $\{(Y_{i,n}, W_{i,n}, X_{i,n})\}_{i=1}^n$ be independent and identically distributed under P_a according to the randomized model

$$Y(w) = m_0(X) + w\tau_a(X) + U, \quad \mathbb{E}(U \mid X, W) = 0, \quad (8)$$

where

$$W \perp \{Y(0), Y(1)\} \mid X, \quad \mathbb{P}(W = 1 \mid X) = \frac{1}{2}, \quad Y = Y(W).$$

Here m_0 is a measurable baseline outcome function and U is the disturbance. The function m_0 and the conditional law of U given (X, W) are held fixed as a varies over $\mathcal{C}_a$. We suppress the triangular-array subscript on observations and write P_X for the X -marginal of P_a . Cross-fitting fold labels, tree subsample indices, honesty allocations, and node-level candidate draws are generated independently of the observations. A fixed representation-independent order resolves any exact criterion ties. A cross-fitted nuisance prediction for observation i excludes the fold containing i . These randomizations are mutually independent except when a common-random-number coupling is explicitly used to compare two marginally identical learner specifications. The assignment probability $e(X) = 1/2$ is known. Write $m_a(X) =$

$\mathbb{E}(Y \mid X) = m_0(X) + \tau_a(X)/2$ and

$$\psi_\theta(Y, W, X; m_a) = (W - 1/2)\{Y - m_a(X) - \theta(W - 1/2)\}.$$

For a parent cell C , the population solution is $\theta_C = \mathbb{E}\{\tau_a(X) \mid X \in C\}$, its Jacobian is $A_C = \mathbb{E}\{(W - e(X))^2 \mid X \in C\} = 1/4$, and the GRF linearized score $\phi_C(Y, W, X) = A_C^{-1}\psi_{\theta_C}(Y, W, X; m_a)$ satisfies

$$\mathbb{E}\{\phi_C(Y, W, X) \mid X = x\} = \tau_a(x) - \theta_C.$$

For $g \in \{1, 2\}$, let $c_n \geq 1$ be an integer. For $\ell = 1, \dots, c_n$, let $\varphi_{g\ell n} : [0, 1] \rightarrow \mathbb{R}$ be a deterministic strictly increasing map with finite values, and set $X_g^{(\ell)} = \varphi_{g\ell n}(X_g)$. The identity map gives an exact copy. Define

$$\begin{aligned} R_{1,n}(X) &= \{X_1^{(1)}, \dots, X_1^{(c_n)}, X_2\}, \\ R_{2,n}(X) &= \{X_1, X_2^{(1)}, \dots, X_2^{(c_n)}\}. \end{aligned}$$

Thus $M_n = c_n + 1$. Strict monotonicity gives the same threshold partitions for every column in a repeated block. In exact arithmetic it also preserves source routing because $\varphi_{g\ell n}(x) \leq \varphi_{g\ell n}(t)$ if and only if $x \leq t$. Proposition 2 treats more general recodings on a declared array under canonical class routing. For every n ,

$$\sigma\{R_{1,n}(X)\} = \sigma\{R_{2,n}(X)\} = \sigma(X_1, X_2).$$

Representation $R_{1,n}$ repeats the weaker modifier X_1 and leaves the stronger modifier X_2 as a singleton. Representation $R_{2,n}$ repeats the stronger modifier and leaves the weaker one as a singleton. The main comparison asks whether a singleton is selected before the target path reaches its leaf. In this signed design, each effect component corresponds one-to-one to a split-action class. We use *component* for the term in the CATE and *class* for the collection of column labels sampled by the tree.

3.2 Source learner and verification objects

Each tree draws candidate columns, evaluates their admissible threshold actions, chooses the largest source criterion, and repeats until the stopping rule is met. An independent honest sample then computes the leaf moment. The learner follows the causal-forest source code in `grf` version 2.4.0 at commit 11ead50b, with unit sample weights, `W.hat=1/2`, `stabilize.splits=TRUE`, and zero imbalance penalty [GRF Development Team, 2024].

Each tree draws s_n observations without replacement and divides them honestly into splitting and estimation samples whose fractions are bounded away from zero. At a node with M representation columns, the source candidate-count rule is

$$N_M \sim \text{Poisson}(\xi_M), \quad Q_M = \max\{1, \min(N_M, M)\}, \quad (9)$$

where $\xi_M = \text{mtry}$ [GRF Development Team, 2026]. Conditional on $Q_M = q$, the implementation samples q columns uniformly without replacement. At each node it evaluates the admissible thresholds of those columns and uses a fixed tie order. For probability calculations, the independent node draws can be indexed in advance by tree and node address. This indexing is an analytical device and leaves the source learner's distribution unchanged. The package continues to use its sequential pseudorandom number generator. Let k_n be the source minimum node size, the required count on each side of the parent treatment mean within each child. Write $m = m_a$, and let $\hat{m}$ be its cross-fitted estimator. We require

$$\|\hat{m} - m\|_\infty = o_p(1), \quad \|\hat{m}\|_\infty = O_p(1) \quad (10)$$

uniformly over $a \in \mathcal{C}_a$. Write

$$\tilde{Y}_i = Y_i - \hat{m}(X_i), \quad Z_i = W_i - 1/2 \in \{-1/2, 1/2\}.$$

For a splitting node C , write $n_C = |C|$, let bars denote unweighted node averages, and set

$$\hat{A}_C = \frac{1}{4} - \bar{Z}_C^2, \quad \hat{\theta}_C = \frac{\overline{Z\tilde{Y}_C} - \bar{Z}_C\bar{\tilde{Y}_C}}{\hat{A}_C}.$$

For a candidate split (j, t) , write $C_L(j, t)$ and $C_R(j, t)$ for its children, and let $\mathcal{Q}_C(j, t)$ denote the corresponding population CART criterion. This criterion is the child-size-weighted squared contrast in mean score between the two children. We use *split gain* for the value of this criterion. Appendix A records the source binding and verifies the exact scalar relabeling formula. Suppress the arguments when clear. For $A \in \{C_L, C_R\}$, write $n_A = |A|$ and $\hat{p}_{A|C} = n_A/n_C$. Under the preceding specialization, the scalar source relabeling and a rescaling of its split criterion are exactly

$$\begin{aligned} \Psi_{i,C} &= (Z_i - \bar{Z}_C)\{\tilde{Y}_i - \bar{\tilde{Y}_C} - \hat{\theta}_C(Z_i - \bar{Z}_C)\}, \\ \hat{\mathcal{Q}}_C^{\text{grf}}(j, t) &= 16\hat{p}_{C_L|C}\hat{p}_{C_R|C}(\bar{\Psi}_{C_L} - \bar{\Psi}_{C_R})^2. \end{aligned} \quad (11)$$

The factor 16 does not affect the source argmax. The source option **alpha** is the minimum information fraction. Write its fixed value as $\beta \in (0, 1/4)$. At node C , let $\mathcal{S}_n(C)$ contain each pair (j, t) such that j is sampled under (9), t is one of the finite source thresholds for column j in C , and both children $A \in \{C_L(j, t), C_R(j, t)\}$ contain at least k_n observations with $Z_i < \bar{Z}_C$ and at least k_n with $Z_i > \bar{Z}_C$, while satisfying $n_A(1/4 - \bar{Z}_A^2) \geq \beta n_C(1/4 - \bar{Z}_C^2)$. The following condition applies to every node reachable under either representation or along either potential failure path in the proof, for every realized nonempty candidate set. We impose

$$\mathbb{P} \left\{ \max_{(j,t) \in \mathcal{S}_n(C)} \hat{\mathcal{Q}}_C^{\text{grf}}(j, t) > 0 \text{ at every such } C \text{ with } \mathcal{S}_n(C) \neq \emptyset \right\} = 1. \quad (12)$$

The probability integrates over the sample, tree subsampling, honesty split, nuisance estimates, outcomes, and candidate randomization. We impose this continuation probability directly because the density assumption below is insufficient by itself. It ensures that the unpenalized source procedure continues whenever an admissible candidate action remains. It is used in the proof to obtain the common path-depth envelope. It places no ordering restriction on competing split gains.

For comparison define the linear orthogonal score

$$\hat{\Gamma}_i = 4(W_i - 1/2)\{Y_i - \hat{m}(X_i)\}.$$

Its CART criterion is

$$\hat{\mathcal{Q}}_C^{\Gamma}(j, t) = \hat{p}_{C_L|C} \hat{p}_{C_R|C} (\bar{\Gamma}_{C_L} - \bar{\Gamma}_{C_R})^2$$

over the same candidate library. Candidate counts and column samples are independent across nodes and independent of the other tree randomization. The estimation half supplies the observations used to compute leaf means, and empty leaves are pruned.

For a forest with B_n trees and an independent target x , let $\alpha_i(x)$ be its nonnegative weights over nonempty honest leaves, with $\sum_i \alpha_i(x) = 1$, and define

$$\bar{Z}_\alpha(x) = \sum_i \alpha_i(x) Z_i, \quad \tilde{Y}_\alpha(x) = \sum_i \alpha_i(x) \tilde{Y}_i, \quad \overline{Z\tilde{Y}}_\alpha(x) = \sum_i \alpha_i(x) Z_i \tilde{Y}_i.$$

The source prediction is a local covariance ratio because the forest solves a weighted treatment-effect moment inside the target's leaves. It is exactly

$$\hat{T}_{n, B_n}(x) = \frac{\overline{Z\tilde{Y}}_\alpha(x) - \bar{Z}_\alpha(x) \tilde{Y}_\alpha(x)}{1/4 - \bar{Z}_\alpha(x)^2}.$$

Its forest-level linear comparator

$$\check{T}_{n,B_n}(x) = \sum_i \alpha_i(x) \hat{\Gamma}_i = 4 \overline{Z \tilde{Y}}_\alpha(x)$$

is the forest average of the generic honest leaf scores $\tilde{T}_{b,n}^R(x)$ defined in Section 2. It satisfies the exact decomposition

$$\hat{T}_{n,B_n}(x) - \check{T}_{n,B_n}(x) = \frac{4 \bar{Z}_\alpha(x) \{4 \overline{Z \tilde{Y}}_\alpha(x) \bar{Z}_\alpha(x) - \tilde{Y}_\alpha(x)\}}{1 - 4 \bar{Z}_\alpha(x)^2}. \quad (13)$$

The package reports $\hat{T}_{n,B_n}$ as its CATE estimate. The theorem below shows that its probability limit can differ from the true CATE when the representation changes candidate access. Let $\hat{T}_{h,n}$ and $\check{T}_{h,n}$, $h \in \{1, 2\}$, denote the source forest and its linear comparator under $R_{h,n}$, each with B_n trees. The production-rate condition is

$$\begin{aligned} s_n \rightarrow \infty, \quad s_n/n \leq \bar{s} < 1, \quad k_n \rightarrow \infty, \quad k_n/s_n \rightarrow 0, \quad k_n/\log(B_n s_n) \rightarrow \infty, \\ B_n \rightarrow \infty, \quad H_n := \log(s_n/k_n) \rightarrow \infty. \end{aligned} \quad (14)$$

The quantity H_n is the effective path depth. A sequence of roughly balanced splits reduces a root sample of order s_n to the minimum leaf scale k_n in order $\log(s_n/k_n)$ steps.

Assumption 3. *Uniformly over $a \in \mathcal{C}_a$, the outcome is bounded, $U \mid X, W$ has a continuous density on a bounded support, the cross-fitted nuisance estimates satisfy (10), and the source nondegeneracy condition (12) holds. Tree groups used for variance estimation have uniformly bounded size, and the sequences (s_n, k_n, B_n) satisfy (14).*

Boundedness and the nuisance rate control the honest score. Split nondegeneracy and the production rates support the leaf and forest approximations. The appendix verifies Assumptions 1 and 2 for the source forest under these conditions.

3.3 Main result: multiplicity and forest limits

Define the unconstrained treatment rules

$$\hat{\pi}_{h,n}(x) = \mathbf{1}\{\hat{T}_{h,n}(x) > 0\}, \quad \pi_a^*(x) = \mathbf{1}\{\tau_a(x) > 0\} = \mathbf{1}\{s_2(x) = 1\}.$$

Let r_g count the representation columns in split-action class g , and set

$$\mathbf{r} = (r_1, \dots, r_G)^\top, \quad M = \sum_{g=1}^G r_g.$$

At node C , class g has finite split-action set $\mathcal{A}_g(C)$, and let $\mathcal{G}(C) = \{g : \mathcal{A}_g(C) \neq \emptyset\}$ be the active classes. For $g \in \mathcal{G}(C)$ and $u \in \mathcal{A}_g(C)$, write $\widehat{\mathcal{Q}}_C(u)$ for the realized source criterion indexed by split action u , and select the class optimum

$$\widehat{u}_g(C) = \arg \max_{u \in \mathcal{A}_g(C)} \widehat{\mathcal{Q}}_C(u)$$

using a fixed deterministic tie rule. Write $\widehat{v}_g(C) = \widehat{\mathcal{Q}}_C\{\widehat{u}_g(C)\}$, let $G_C = |\mathcal{G}(C)|$, and let $g_j(C)$ denote the active class with the j th largest realized score. A fixed representation-independent class order breaks ties, so

$$\widehat{v}_{g_1(C)}(C) \geq \dots \geq \widehat{v}_{g_{G_C}(C)}(C).$$

For a general candidate-count rule, let $\nu_M(q) = \mathbb{P}(Q_M = q)$ on $\{1, \dots, M\}$. Conditional on $Q_M = q$, sample q columns uniformly without replacement. For the class ranking defined above, define cumulative multiplicity by

$$M_{j,C}^{\text{cum}} = \sum_{\ell=1}^j r_{g_\ell(C)}, \quad M_{0,C}^{\text{cum}} = 0.$$

Conditional on the realized node state and this score order, the probability that active class $g_j(C)$, $j = 1, \dots, G_C$, is chosen for the split is

$$w_{j,C}(\mathbf{r}; q) = \frac{\binom{M - M_{j-1,C}^{\text{cum}}}{q} - \binom{M - M_{j,C}^{\text{cum}}}{q}}{\binom{M}{q}}, \quad (15)$$

where $\binom{a}{q} = 0$ for $a < q$. Under candidate-count law ν_M ,

$$w_{j,C}(\mathbf{r}; \nu_M) = \sum_{q=1}^M \nu_M(q) w_{j,C}(\mathbf{r}; q), \quad (16)$$

and class g is available with probability

$$\omega_g(\mathbf{r}; \nu_M) = \sum_{q=1}^M \nu_M(q) \left\{ 1 - \frac{\binom{M-r_g}{q}}{\binom{M}{q}} \right\}. \quad (17)$$

The winner probabilities need not sum to one when a candidate draw contains only inactive classes. The remaining mass is the probability that the node has no admissible sampled action.

Set

$$\omega_n = \frac{\mathbb{E}Q_{M_n}}{M_n}, \quad \mathcal{I}_n = H_n \omega_n.$$

Call ω_n the singleton's per-node inclusion probability and $\mathcal{I}_n$ its cumulative-inclusion index. For $h \in \{1, 2\}$, write $\bar{h} = 3 - h$. In this design, success at a reached node means that the chosen split uses singleton $X_{\bar{h}}$. It is a property of the node transition and does not depend on which evaluation point later traverses the node. Conditional on the training state, construct a full potential failure tree $\mathcal{T}_{h,n}^0$ recursively. At each node, draw the source transition from its regular conditional law given that the chosen split does not use $X_{\bar{h}}$, and generate both children with independent node-address randomizers. Null conditional branches are absorbing. The evaluation point $X \sim P_X$ is drawn independently after this tree is built and routed through it. This construction realizes the failure kernels in Section 2 while retaining the original M_n -column candidate law, stopping rule, and competing actions. No column is deleted. Let $J_{h,n}^0(x)$ be the number of internal nodes on the resulting path to the terminal leaf. Let $\rho_{h,n,t}^0(x)$ be the history-conditional hazard that the singleton is selected and resolved at internal node t , and set

$$\Lambda_{h,n}^0(x) = \sum_{t=1}^{J_{h,n}^0(x)} \rho_{h,n,t}^0(x).$$

The leading index h identifies representation $R_{h,n}$, while the unresolved component is $\bar{h}$. In the abstract notation, $S_{\bar{h},n,t}^{R_{h,n}}(x) = 1$ exactly when the chosen split uses $X_{\bar{h}}$, and

$$J_{h,n}^0 = J_{\bar{h},n}^{R_{h,n},0}, \quad \rho_{h,n,t}^0 = \rho_{\bar{h},n,t}^{R_{h,n},0}, \quad \Lambda_{h,n}^0 = \Lambda_{\bar{h},n}^{R_{h,n}}.$$

Appendix A establishes threshold localization.

The counterfactual path evolves under repeated singleton failure and need not coincide with an observed tree path. The index $\mathcal{I}_n = H_n \omega_n$ combines its two relevant scales. The path contains order- H_n split nodes, and the singleton enters each candidate set with probability ω_n . Split competition determines whether candidate inclusion resolves it. The proof estab-

lishes the source-score transfer, honest reduction, depth bounds, and hazard approximation. The intermediate formula additionally requires normalized path depth to converge.

For a fixed $a \in \mathcal{C}_a$ and a sequence with $\mathcal{I}_n \rightarrow \kappa \in (0, \infty)$, the intermediate regime additionally requires the pointwise depth condition under the joint law $\mathbb{P}_a^0$ of the training sample, the full failure tree, and an independent $X \sim P_X$ routed through that tree,

$$\frac{J_{h,n}^0(X)}{H_n} \xrightarrow[\mathbb{P}_a^0]{p} d_h(a) \in (0, \infty), \quad h = 1, 2, \quad (18)$$

and we write $\zeta_h(a) = \exp\{-\kappa d_h(a)\}$.

Theorem 2 (Limits of causal-forest predictions). *Under Assumption 3, the following two endpoint results hold uniformly over $a \in \mathcal{C}_a$. If $\mathcal{I}_n \rightarrow 0$, then*

$$\max_{h \in \{1,2\}} \|\widehat{T}_{h,n} - a_h s_h\|_{L^1(P_X)} \xrightarrow{p} 0. \quad (19)$$

If $\mathcal{I}_n \rightarrow \infty$, then

$$\max_{h \in \{1,2\}} \|\widehat{T}_{h,n} - \tau_a\|_{L^1(P_X)} \xrightarrow{p} 0. \quad (20)$$

For any fixed $a \in \mathcal{C}_a$, if $\mathcal{I}_n \rightarrow \kappa \in (0, \infty)$ and (18) holds under $\mathbb{P}_a^0$, then

$$\widehat{T}_{1,n} \xrightarrow{p} T_{1,\kappa} = a_1 s_1 + \{1 - \zeta_1(a)\} a_2 s_2, \quad (21)$$

$$\widehat{T}_{2,n} \xrightarrow{p} T_{2,\kappa} = \{1 - \zeta_2(a)\} a_1 s_1 + a_2 s_2 \quad (22)$$

in $L^1(P_X)$.

The product $H_n \omega_n$ gives the expected order of singleton appearances along the path. If it tends to zero, the singleton usually never enters the search and its component drops out. If it diverges, the singleton is offered often enough for recovery. At an interior limit, the depth condition (18) determines the remaining attenuation. The two endpoint results hold without this additional condition. Across all three cases, the representations differ only in how candidate access is allocated among equivalent columns.

For a treatment rule π , define

$$V_a(\pi) = \mathbb{E}\{\tau_a(X)\pi(X)\}, \quad \text{Reg}_a(\pi) = V_a(\pi_a^*) - V_a(\pi).$$

The regret identity

$$\text{Reg}_a(\pi) = \mathbb{E}[|\tau_a(X)| \mathbf{1}\{\pi(X) \neq \pi_a^*(X)\}] \quad (23)$$

shows that a wrong sign is a treatment mistake weighted by its causal margin.

Corollary 1 (Sign reversal and policy regret). *Under the conditions of Theorem 2, if $\mathcal{I}_n \rightarrow 0$, then*

$$\left| \mathbb{P}_X \{ \widehat{T}_{1,n}(X) \widehat{T}_{2,n}(X) < 0 \} - \frac{1}{2} \right| \rightarrow_p 0, \quad (24)$$

and the reversal is separated from zero.

$$\mathbb{P}_X \left\{ \begin{array}{l} \widehat{T}_{1,n}(X) \widehat{T}_{2,n}(X) < 0, \\ \min(|\widehat{T}_{1,n}(X)|, |\widehat{T}_{2,n}(X)|) \geq \underline{a}/2, \\ |\tau_a(X)| \geq \Delta_a \end{array} \right\} \rightarrow_p \frac{1}{2}. \quad (25)$$

Moreover,

$$\text{Reg}_a(\widehat{\pi}_{2,n}) \rightarrow_p 0, \quad (26)$$

$$\left| \text{Reg}_a(\widehat{\pi}_{1,n}) - \frac{a_2 - a_1}{2} \right| \rightarrow_p 0. \quad (27)$$

Because $a_2 > a_1$, the representation that repeats X_2 yields the optimal unconstrained rule in this regime. Repeating X_1 instead produces limiting regret $(a_2 - a_1)/2$. The corollary concerns unconstrained treatment assignment. With a capacity constraint, selection depends on ranks around an endogenous cutoff. Section 5 examines the sensitivity of such a ranking.

3.4 Finite-dimensional extension

For a fixed $G \geq 2$, let

$$X = (X_1, \dots, X_G) \sim \text{Unif}([0, 1]^G), \quad s_g = 2\mathbf{1}\{X_g \geq 1/2\} - 1.$$

Define

$$\mathcal{C}_G = \left\{ a \in [\underline{a}_G, \bar{a}_G]^G : \min_{g \neq h} |a_g^2 - a_h^2| \geq \Delta_G \right\}, \quad \tau_a^{(G)} = \sum_{g=1}^G a_g s_g, \quad (28)$$

where $0 < \underline{a}_G < \bar{a}_G < \infty$ and $0 < \Delta_G < (\bar{a}_G^2 - \underline{a}_G^2)/(G-1)$, so $\mathcal{C}_G$ has nonempty interior. For each g , let $R_{g,n}^{(G)}$ contain $c_n \geq 1$ deterministic strictly increasing recodings of X_g , including the identity, and one copy of every X_h , $h \neq g$. Then $M_{G,n} = c_n + G - 1$, and every representation generates $\sigma(X)$. The corresponding per-node inclusion probability and cumulative-inclusion index are

$$\omega_{G,n} = \frac{\mathbb{E} Q_{M_{G,n}}}{M_{G,n}}, \quad \mathcal{I}_{G,n} = H_n \omega_{G,n}.$$

Run the same source learner and bounded randomized model with target $\tau_a^{(G)}$, under the corresponding fixed- G version of Assumption 3. Write $\widehat{T}_{g,n}^{(G)}$ and $\tilde{T}_{g,n}^{(G)}$ for the resulting prediction and linear comparator.

Proposition 1 (Fixed-dimensional component selection). *Uniformly over $a \in \mathcal{C}_G$,*

$$\mathcal{I}_{G,n} \rightarrow 0 \implies \max_{1 \leq g \leq G} \|\widehat{T}_{g,n}^{(G)} - a_g s_g\|_{L^1(P_X)} \xrightarrow{p} 0, \quad (29)$$

$$\mathcal{I}_{G,n} \rightarrow \infty \implies \max_{1 \leq g \leq G} \|\widehat{T}_{g,n}^{(G)} - \tau_a^{(G)}\|_{L^1(P_X)} \xrightarrow{p} 0. \quad (30)$$

For every $g \neq h$, the first regime also gives

$$\left| \mathbb{P}_X\{\widehat{T}_{g,n}^{(G)}(X)\widehat{T}_{h,n}^{(G)}(X) < 0\} - \frac{1}{2} \right| \xrightarrow{p} 0. \quad (31)$$

Proposition 1 extends the mechanism from two effect modifiers to any fixed number. When cumulative singleton inclusion vanishes, the representation that repeats X_g selects the single component $a_g s_g$. The G information-equivalent representations can therefore converge to G different component targets, and any pair disagrees in sign on half the covariate space. When cumulative inclusion diverges, every representation recovers the full target $\tau_a^{(G)}$. The proposition keeps G fixed and does not cover a growing number of effect components.

Corollary 2 (Multiplicity boundary). *Under Assumption 3, for the source learner above, if $M_n \rightarrow \infty$ and $\xi_{M_n} \equiv \xi$ for a fixed $\xi > 0$, then*

$$\omega_n \sim \frac{\xi + e^{-\xi}}{M_n}, \quad (32)$$

so the critical multiplicity is $M_n \asymp H_n$. Under package-default `mtry` scaling

$$\xi_M = \min\{\lceil \sqrt{M} \rceil + 20, M\}, \quad (33)$$

$\omega_n \sim M_n^{-1/2}$, and the critical multiplicity is $M_n \asymp H_n^2$. Consequently,

$$\frac{M_n}{H_n^2} \rightarrow \infty \implies \mathcal{I}_n \rightarrow 0, \quad \frac{M_n}{H_n^2} \rightarrow 0 \implies \mathcal{I}_n \rightarrow \infty. \quad (34)$$

The same boundaries hold with $M_{G,n}$ and $\mathcal{I}_{G,n}$. Under fixed `mtry`, $M_{G,n} \asymp H_n$, whereas under (33), $M_{G,n} \asymp H_n^2$. Every fixed finite representation is in the recovery regime.

Corollary 2 translates resolution intensity into a comparison between path depth and raw dimension. With fixed `mtry`, a singleton enters a node's candidate set with probability

of order M_n^{-1} , giving order H_n/M_n inclusions along a target path. Under package-default scaling, this count is of order $H_n/\sqrt{M_n}$. Above the corresponding multiplicity scale, the singleton is unlikely to enter a candidate set before the leaf is reached. Below that scale, repeated inclusion permits recovery.

The phrase package default refers only to the `mtry` rule in (33). The production sequence in (14) continues to require $k_n \rightarrow \infty$ and therefore differs from repeated fits that hold the package’s finite-sample node-size setting fixed.

These boundaries compare sequences of representations as the sample size grows. Equation (24) concerns the infinite-forest target. Finite-ensemble simulation noise is separate. In particular, (34) cannot produce inconsistent limits from two fixed finite representations.

4 Finite-Array Class Sampling

Ordinary `grf` samples raw columns. This section defines a new learner that samples certified split classes. Group-first sampling also appears in module-guided forests, where correlation networks define the groups [Chen and Zhang, 2013]. Here, high correlation is insufficient because two variables may encode different economic states. The correction begins with a declared finite array

$$\mathcal{X}_{\text{cert}} = \{x_1, \dots, x_N\}$$

that contains every point used for splitting, nuisance fitting, prediction, variance estimation, and policy ranking. The array is complete, with no missing covariate entries. A column’s certificate is its weak ordering of $\mathcal{X}_{\text{cert}}$, identified with the reversed ordering. Equal certificates generate the same unordered threshold bipartitions on every subset of $\mathcal{X}_{\text{cert}}$, and the certificates can be computed before the forest is grown.

Let K denote the number of certified classes. The class-sampled forest has three steps. First, compute each column’s integer weak-rank vector, retain the lexicographically smaller of its forward and reverse strings, and map every class to a canonical rank column. The implementation uses exact certificate comparison in place of a probabilistic hash. Second, draw $\tilde{Q}_K$ classes uniformly without replacement and evaluate each distinct action once, where $\tilde{\nu}_K(q) = \mathbb{P}(\tilde{Q}_K = q)$ for $q = 1, \dots, K$. Third, index child orientation, ties, and random numbers by the ordered certificate. These operations make the certified split class the unit of randomization. Class reduction occurs before the candidate draw, so candidates are sampled from K classes. Deleting duplicate actions afterward would retain the ordinary column-sampling probabilities.

For an active certified class $g \in \mathcal{G}(C)$, let $h_g(C)$ be the number of active certified classes

ranked ahead of g by the score and fixed tie rule. A sampled inactive class contributes no admissible action and does not block an active class. The class-level winning probability is

$$\tilde{p}_g(C) = \sum_{q=1}^K \tilde{\nu}_K(q) \frac{\binom{K-h_g(C)}{q} - \binom{K-h_g(C)-1}{q}}{\binom{K}{q}}. \quad (35)$$

For example, a raw variable, its logarithm, and a reversed unit conversion have one certificate whenever they induce the same weak ordering and tie blocks on the declared array. The class-sampled forest represents that certificate once and samples from the reduced class matrix.

Fix $\mathcal{X}_{\text{cert}}$, the certification rule, and the class-level candidate-count law $\tilde{\nu}_K$. Replace the raw array by its canonical class matrix before fitting. The coupled learners receive the same outcomes, treatments, weights, cluster and unit identifiers, policy capacity, and tie keys. They also use the same class-indexed folds, tree randomizers, variance groups, and policy-ranking randomizers. Every downstream operation is assumed to be a measurable function of these common inputs.

Proposition 2 (Class-sampled forest). *Under the common class-level coupling just described, any operation preserving the certified class set and its canonical routing leaves the restriction of every tree partition to $\mathcal{X}_{\text{cert}}$, every forest weight, prediction, reported standard error, and selected set on $\mathcal{X}_{\text{cert}}$ unchanged. This includes permutation, insertion or deletion within a represented class, complementation, and recoding that preserves or reverses the observed weak order and ties. Winning probabilities are given by (35) and do not depend on raw within-class multiplicities.*

Equality of weak sample orderings certifies threshold partitions on a fixed finite array. If $\mathcal{X}_{\text{cert}}$ includes the evaluation points, the predictions and policy-selected sets are exactly invariant there. The implementation maps each class to a canonical weak-rank coordinate and orders classes by certificate before calling the forest engine. Only exact weak-order classes are combined. Correlated variables with different orderings remain separate, and the certification supplies no guarantee at undeclared points. Under a common seed, two certified representations pass identical ordered class matrices and the same candidate-count parameter to the engine. The zero differences reported below verify this algorithmic equality. Predictive accuracy and the validity of reported variance estimates remain separate questions. In the empirical checks, $\tilde{Q}_K$ uses the capped-Poisson rule in (9), with the same numerical `mtry` mean as the corresponding baseline fit and with the draw capped at K .

5 Finite-Sample Evidence

5.1 Monte Carlo

The Monte Carlo measures whether equivalent representations imply opposite treatment rules for the same individual and compares those disagreements with ordinary seed variation. The theorem uses strictly increasing recodings, bounded outcomes, and asymptotic learner sequences. The experiment also uses order-reversing and rank recodings with Gaussian outcomes in finite samples. Draw $X \sim \text{Unif}([0, 1]^p)$ for $p \geq 6$, $W \sim \text{Bernoulli}(1/2)$, and let $s_j = 2\mathbf{1}\{X_j > 1/2\} - 1$. The treatment effect and outcome are

$$\begin{aligned}\tau(X) &= 0.20 + 0.65s_1 - 0.55s_2 + 0.10(X_3 - 0.5), \\ Y &= 1 + X_4 + 0.5X_5^2 - 0.5X_6 + W\tau(X) + \varepsilon, \quad \varepsilon \sim N(0, 1).\end{aligned}$$

The remaining $p - 6$ coordinates are independent noise covariates. For $j \in \{1, 2\}$, representation $R_\ell^{(j)}$ contains X and ℓ added order-preserving or order-reversing encodings of X_j . The library contains changes of units, complements, logs, ranks, powers, and smooth monotone maps. The replication README records the ordered library. On the pooled training-and-evaluation array, $R_\ell^{(1)}$ and $R_\ell^{(2)}$ generate the same information and the same distinct threshold partitions. They differ only in which effect modifier has more column labels in candidate sampling. None of these added encodings is numerically identical to its source column. Pooling is used only to construct the covariate recodings consistently on this finite array. Because every map is strictly monotone or order reversing on its source column, it preserves all distinct finite-array thresholds. The construction concerns only this observed array. Future observations require a separately specified preprocessing rule.

For evaluation observation i , let $\hat{\tau}_{j,\ell,i} = \hat{\tau}_{j,\ell}(X_i)$ be the prediction under $R_\ell^{(j)}$, and write $\mathbb{E}_n h_i = N_{\text{eval}}^{-1} \sum_{i=1}^{N_{\text{eval}}} h_i$. The implied unconstrained rule is $\hat{\pi}_{j,\ell}(X_i) = \mathbf{1}\{\hat{\tau}_{j,\ell,i} > 0\}$. We report three paired disagreement rates. Each compares instability between the two representations. The experiment does not estimate bias, mean squared error, or policy value.

$$\begin{aligned}\text{SR}_0 &= \mathbb{E}_n \mathbf{1}\{\hat{\tau}_{1,\ell,i} \hat{\tau}_{2,\ell,i} < 0\}, \\ \text{SR}_{0.10} &= \mathbb{E}_n \mathbf{1}\{\hat{\tau}_{1,\ell,i} \hat{\tau}_{2,\ell,i} < 0, \min(|\hat{\tau}_{1,\ell,i}|, |\hat{\tau}_{2,\ell,i}|) \geq 0.10\}, \\ \text{SR}_{\tau,0.25} &= \frac{\mathbb{E}_n \mathbf{1}\{\hat{\tau}_{1,\ell,i} \hat{\tau}_{2,\ell,i} < 0, |\tau(X_i)| \geq 0.25\}}{\mathbb{E}_n \mathbf{1}\{|\tau(X_i)| \geq 0.25\}}.\end{aligned}$$

Every cell averages twenty independently generated training samples of size 3,000, each paired with an independent evaluation sample of size 30,000. Forests use `grf` 2.4.0, 2,000

trees, oracle nuisance functions, and common data and forest seeds across representations. Each replication is therefore paired across representations. Their different column counts prevent a common seed from aligning every internal random-number stream.

Panel A uses the package-default search budget. With eight added encodings, overall sign disagreement rises from 6.83 percent at baseline dimension 40 to 46.22 percent at dimension 500. At the low dimension, nearly all reversals occur close to zero. At dimension 500, 31.37 percent of predictions disagree after both estimates are required to exceed 0.10 in absolute value, and 28.99 percent disagree among observations with $|\tau(X)| \geq 0.25$. At high dimension, the discrepancy is no longer confined to predictions near zero.

Panel B holds the baseline dimension and the `mtry` Poisson mean fixed, which isolates the role of multiplicity. Raising the number of added encodings from one to sixteen increases overall sign disagreement from 17.69 to 41.33 percent and $\text{SR}_{0.10}$ from 4.00 to 29.50 percent. This increase is the finite-sample counterpart of the singleton’s declining inclusion probability as the repeated class grows.

Panel C separates representation dependence from ordinary forest randomness. At $p = 500$, changing the representation produces 46.10 percent sign disagreement. Changing only the seed produces 0.09 percent. The corresponding 95 percent t_{19} intervals for the Monte Carlo mean across twenty replications are [44.05, 48.15] and [0.051, 0.120] percent. The tree-count rows of Table 3 are also stable between 800 and 5,000 trees. Quotienting the certified split classes reduces every paired discrepancy to zero. In this design, the effect of the representation is much larger than seed variation and remains after additional tree averaging. The exercise quantifies the effect for this design. Its frequency in applied work remains unknown.

Table 1: Estimated-CATE sign disagreement under equivalent representations

Comparison	Raw dim.	SR ₀ (%)	SR _{0.10} (%)	SR _{$\tau, 0.25$} (%)
<i>Panel A: package-default search, $\ell = 8$</i>				
$p = 40$	48	6.83 (1.24)	0.02 (0.01)	0.40 (0.15)
$p = 100$	108	21.86 (1.30)	10.57 (1.82)	2.50 (0.75)
$p = 250$	258	37.23 (1.98)	25.96 (1.51)	17.26 (2.77)
$p = 500$	508	46.22 (0.99)	31.37 (2.10)	28.99 (1.33)
<i>Panel B: fixed Poisson mean, $p = 40$, $mtry = 6$</i>				
$\ell = 1$	41	17.69 (1.42)	4.00 (0.85)	1.06 (0.33)
$\ell = 2$	42	24.50 (1.15)	13.38 (1.64)	3.72 (1.50)
$\ell = 4$	44	30.06 (1.59)	21.00 (1.02)	8.56 (2.17)
$\ell = 8$	48	35.54 (1.79)	25.54 (1.25)	15.28 (2.41)
$\ell = 16$	56	41.33 (1.65)	29.50 (1.58)	22.76 (2.20)
<i>Panel C: diagnostics at $p = 500$, $\ell = 8$, package default</i>				
Representation change	508	46.10 (0.98)	31.39 (2.10)	28.83 (1.32)
Second forest seed	508	0.09 (0.02)	0.00 (0.00)	0.02 (0.00)
Class-sampled forest	508	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)

Notes: Entries are Monte Carlo means with standard errors across twenty replications in parentheses. SR₀ is the share of evaluation observations with opposite-signed paired predictions. SR_{0.10} additionally requires both predictions to have absolute magnitude at least 0.10. SR _{$\tau, 0.25$} conditions on an absolute true treatment effect of at least 0.25. Here ℓ is the number of added encodings, so the enriched split class contains $\ell + 1$ raw columns and total raw dimension is $p + \ell$. Panel C is an independent benchmark using the design in the last row of Panel A. The second-seed row changes only the forest seed of $R_8^{(1)}$. The class-sampled row compares the representations after quotienting certified split classes. Each entry compares $R_\ell^{(1)}$ with $R_\ell^{(2)}$, not either enriched representation with the raw baseline.

5.2 Cash-transfer ranking sensitivity

The Kenya exercise asks how a hypothetical top-half ranking changes when baseline covariates are recoded without changing their information. I fit the forests in sample and rank households using out-of-bag predictions. Because the experimental design includes saturation and interference, I interpret the switching rates only as descriptive measures of ranking sensitivity. The data are the public household files from the randomized cash-transfer experiment in Kenya analyzed by Egger et al. [2022]. The outcome is endline per-capita consumption, treatment is the randomized transfer indicator, and the sample contains 4,765 eligible households with baseline data in 653 villages. Sixteen baseline predictors describe household composition, labor-market status, land and livestock ownership, durable ownership, meals, and protein meals.

The outcome is the released winsorized PPP measure of per-capita consumption. Continuous baseline variables are median-imputed and indicator variables are zero-imputed. A weighted regression forest supplies the common outcome nuisance estimate, and the weighted treatment mean supplies the propensity nuisance. The replication README lists the exact predictors and recodings.

Every fit uses **grf** version 2.4.0, 2,000 trees, village clusters, survey weights, a common outcome nuisance estimate, the same seed, and **mtry** = 16. In this implementation, **mtry**

is the mean ξ_M of the uncapped Poisson draw N_M in (9). The source then clamps that draw to obtain the candidate count Q_M . Holding this mean fixed isolates representation multiplicity and differs from recomputing the package default after each augmentation. I rank households by their out-of-bag causal-forest treatment-effect scores and select the top half, with household identifiers breaking numerical ties. Let S_0 be the selected set under the canonical representation and S_R the selected set under recoding R . The reported switching rate is

$$\text{Switch}(R) = \frac{1}{N} \sum_{i=1}^N |\mathbf{1}\{i \in S_R\} - \mathbf{1}\{i \in S_0\}| = \frac{|S_R \triangle S_0|}{N}. \quad (36)$$

It is the fraction of households whose selected status changes and counts both exits and entries. Each set contains exactly $\lfloor N/2 \rfloor$ households.

Table 2 separates low-order recodings, concentrated stress tests, and a balanced-multiplicity control. Table 3 collects the tree-count and quotient checks discussed below.

Panel A shows that small, familiar recodings can matter more than the forest seed. A second household-size label switches 5.25 percent of top-half selections, and retaining raw, log, and rank versions of three ordered predictors switches 6.55 percent. The largest change across three alternative seeds is 0.55 percent.

Panel B deliberately concentrates multiplicity on one predictor at a time. Selection switching ranges from 12.76 percent with 32 household-size copies to 25.18 percent with 128 copies of the female-head indicator. The candidate-count law quantifies the induced change in inclusion probabilities. In the canonical $M = 16$ representation, $\mathbb{E}Q_M = 14.41$ and each class is included with probability 0.9008. With 128 copies of household size and fifteen singletons, $M = 143$, the repeated class is included with probability above 0.999999 and each singleton with probability 0.1119. These probabilities describe how strongly the representation changes access to split search. Conditional split scores then determine which included class wins.

Panel C multiplies every predictor equally. This preserves relative class multiplicity, but a candidate set can contain several labels for the same class and consequently fewer distinct split actions. The 6.67 percent switch rate shows that balanced duplication can change search breadth even without favoring a particular predictor.

In the Monte Carlo, $\text{SR}_{0.10}$, the share of observations with opposite signs and both predictions at least 0.10 in absolute value, is 25.67 percent with 800 trees and 25.76 percent with 5,000 trees. Across the Kenya recoding bundles, the ordinary forest changes selected status for 5.86 to 7.54 percent of households on average, while the class-sampled forest records zero changes for every bundle and seed. The finite-array check also gives identical predictions and reported standard errors. Increasing the tree count leaves the Monte Carlo sensitivity essen-

Table 2: Representation sensitivity in the cash-transfer data

Representation	Raw columns	Spearman correlation	Selected status changed (%)
<i>Panel A: seed benchmark and low-order recodings</i>			
Different seed, maximum	16	0.9999	0.55
Household size $\times 2$	17	0.984	5.25
Raw, log, and rank	22	0.978	6.55
<i>Panel B: concentrated multiplicity stress tests</i>			
Household size $\times 32$	47	0.902	12.76
Household size $\times 128$	143	0.751	21.32
Female head $\times 128$	143	0.693	25.18
Employed $\times 128$	143	0.824	18.38
Protein meals $\times 128$	143	0.711	23.17
<i>Panel C: balanced multiplicity control</i>			
Balanced $\times 8$	128	0.980	6.67

Notes: The canonical specification has sixteen columns. All additional recodings in the table were constructed for this diagnostic. Spearman correlation and $\text{Switch}(R)$ in (36) compare each out-of-bag forest-score ranking with the canonical fit under seed 20260806. The first row reports the maximum change across three alternative canonical seeds. Observations, nuisances, tree count, sample weights, clusters, and `mtry` are held fixed. A label of $\times k$ means that the named variable has k total copies. The balanced design gives every canonical predictor eight total copies.

tially unchanged. Class-level sampling eliminates the paired discrepancies on the certified arrays.

Table 3: Robustness and invariance checks

Diagnostic	Raw design	Ordinary forest	Class-sampled forest
<i>Panel A: tree-count robustness, fixed <code>mtry</code> mean 6, $SR_{0.10}$ (%)</i>			
800 trees	$p = 40, \ell = 8$	25.67 (1.25)	—
5,000 trees	$p = 40, \ell = 8$	25.76 (1.29)	—
<i>Panel B: Kenya selected-status switches (%)</i>			
Administrative aliases	20 columns	6.18	0.00
Unit recodings	19 columns	5.86	0.00
Binary complements	19 columns	6.28	0.00
Raw, log, and rank	22 columns	6.53	0.00
Combined routine bundle	32 columns	7.54	0.00
<i>Panel C: finite-array output equality</i>			
Maximum prediction difference	3 versus 7 columns, $K = 3$	—	0
Maximum standard-error difference	3 versus 7 columns, $K = 3$	—	0

Notes: Panel A reports Monte Carlo means with standard errors across twenty replications in parentheses with the `mtry` Poisson mean fixed at 6. Here `mtry` is the uncapped Poisson mean ξ_M . The realized candidate count Q_M is capped at the raw dimension. Panel B reports the ordinary forest’s mean $\text{Switch}(R)$ across three seeds and the maximum value for the class-sampled forest across those seeds. Panel C is a deterministic finite-array check with 400 training observations, 200 evaluation observations, 400 trees, a common seed, and `grf` 2.4.0. Entries are maximum absolute differences over the evaluation array. The canonical and augmented inputs have three and seven raw columns, respectively, and both reduce to the same three certified classes.

6 Conclusion

Feature subsampling makes raw column labels part of a forest’s specification. Two representations can contain the same covariate information and still assign different probabilities to the same split actions. In the signed `grf` design, cumulative candidate inclusion determines each effect component’s weight in the limiting target.

The inconsistent limits require multiplicity to grow with the sample. Every fixed finite representation covered by the theorem recovers under its conditions. Finite samples can nevertheless display large, well-separated sign changes. In the Kenya diagnostic, author-constructed preprocessing-style recodings change selected status for roughly 6 to 8 percent of households, compared with a maximum of 0.55 percent among three alternative seeds.

Practitioners should treat equivalent encodings as part of the forest design and report sensitivity to them. The class-sampled construction gives exact invariance on any array where the split classes have been certified. Applying the same guarantee to future points requires a defensible global routing rule, and learning economically valid classes remains open.

Data and Code Availability

The paper, source files, code, included outputs, and verification scripts are available at <https://github.com/YiNiu-bot/representation-multiplicity-causal-forests>. The Kenya diagnostic uses the public files accompanying Egger et al. [2022], which are not redistributed. The repository records their checksums and maps each reported result to portable reproduction code and an included output.

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A Proofs and Source Verification

Proof of Theorem 1. Fix a representation R and component g , and condition throughout on $(\mathcal{D}_n, \mathcal{Z}_n)$. Put $\mathcal{A}_n^\Theta = \sigma(\mathcal{D}_n, \Theta, \mathcal{Z}_n)$. Uniformly in the integrated norm, the conditional score decomposition gives

$$\mathbb{E}\{\Gamma_n(Y', W', X') \mid X' \in L_n^R(x), \mathcal{A}_n^\Theta\} = \mu + \sum_{h=1}^G \mathbb{E}\{f_h(X') \mid X' \in L_n^R(x), \mathcal{A}_n^\Theta\} + o_p(1).$$

Combining this identity with (2) and (3) gives the component leaf reduction in (6).

Let $X \sim P_X$ be independent of the training sample and draw a fresh potential failure path conditional on $(X, \mathcal{D}_n, \mathcal{Z}_n)$. Write

$$P_{g,n}^R(X) = \prod_{t=1}^{J_{g,n}^{R,0}(X)} \{1 - \rho_{g,n,t}^{R,0}(X)\}.$$

Equation (1) follows from the chain rule for the sequential conditional failure kernels. Entry into the candidate set alone does not determine success, and no threshold-localization approximation enters this identity. If a reached history has hazard one, its factor is zero and the path moves to the absorbing state, so the product still equals the zero conditional failure probability.

Write $\Lambda = \Lambda_{g,n}^R(X)$ and $\bar{\rho} = \bar{\rho}_{g,n}^R(X)$. On $\{\bar{\rho} < 1\}$,

$$\begin{aligned}
0 &\leq -\log P_{g,n}^R(X) - \Lambda \\
&= \sum_{t=1}^{J_{g,n}^{R,0}(X)} \{-\log(1 - \rho_{g,n,t}^{R,0}(X)) - \rho_{g,n,t}^{R,0}(X)\} \\
&\leq \frac{\bar{\rho}}{1 - \bar{\rho}} \Lambda.
\end{aligned} \tag{37}$$

On $\{\lambda_g^R(X) < \infty\}$, conditions (5) and (37) imply $P_{g,n}^R(X) \rightarrow_p e^{-\lambda_g^R(X)}$. On $\{\lambda_g^R(X) = \infty\}$,

$$0 \leq P_{g,n}^R(X) \leq e^{-\Lambda_{g,n}^R(X)} \rightarrow_p 0.$$

Thus, under the joint law of X and the potential path, $P_{g,n}^R(X) - e^{-\lambda_g^R(X)} \rightarrow_p 0$. The difference is bounded, so it also converges to zero in mean. The failure-path identity and Jensen's inequality therefore give

$$\mathbb{E} \int \left| \mathbb{E}_{\Theta} \{1 - D_{g,n}^R(x) \mid \mathcal{D}_n, \mathcal{Z}_n\} - e^{-\lambda_g^R(x)} \right| dP_X(x) \leq \mathbb{E} |P_{g,n}^R(X) - e^{-\lambda_g^R(X)}| \rightarrow 0.$$

Markov's inequality therefore gives

$$\int \left| \mathbb{E}_{\Theta} \{1 - D_{g,n}^R(x) \mid \mathcal{D}_n, \mathcal{Z}_n\} - e^{-\lambda_g^R(x)} \right| dP_X(x) = o_p(1). \tag{38}$$

Because g was arbitrary, (38) holds for every $h \in \{1, \dots, G\}$. Let $\bar{T}_n^R(x) = \mathbb{E}_{\Theta} \{\tilde{T}_n^R(x) \mid \mathcal{D}_n, \mathcal{Z}_n\}$. By the component leaf reduction, Jensen's inequality, and (38),

$$\left\| \bar{T}_n^R - \mu - \sum_{h=1}^G \{1 - e^{-\lambda_h^R(\cdot)}\} f_h \right\|_{L^1(P_X)} = o_p(1). \tag{39}$$

Conditional on $(\mathcal{D}_n, \mathcal{Z}_n)$, partition the trees into mutually independent groups of size at most $L < \infty$, with $L = 1$ in the conditionally independent case. Conditional covariances vanish across groups. Within a group, Cauchy-Schwarz and the bounded second moment bound every covariance by a constant. Hence

$$\mathbb{E} \left[\left\| \frac{1}{B_n} \sum_{b=1}^{B_n} \tilde{T}_{b,n}^R - \bar{T}_n^R \right\|_{L^2(P_X)}^2 \middle| \mathcal{D}_n, \mathcal{Z}_n \right] \leq \frac{CL}{B_n} = o(1).$$

The L^1 -norm is bounded by the L^2 -norm. Combining this fact with (4) and (39) proves (7).

The endpoint statements follow when the limit function $\lambda_g^R(\cdot)$ is identically zero, finite and positive, or infinite. $\square$

Proof of Theorem 2. All probability statements below are uniform over $a \in \mathcal{C}_a$. Constants may depend on $(\underline{a}, \bar{a}, \Delta_a, \beta)$, but not on a or n . Write $\mathcal{D}_n^+ = \sigma(\mathcal{D}_n, \mathcal{Z}_n)$. Conditional source-proof statements hold the nuisance-training folds, fitted nuisance functions, and their training randomization fixed by conditioning on $\mathcal{D}_n^+$.

For every tree root, generate the subsample and honesty allocation first. The ordinary tree uses independent uniforms indexed by node address to realize the source transition law. For each representation $h \in \{1, 2\}$, construct a full failure tree $\mathcal{T}_{h,n}^0$ on the same persistent root samples. At a reached node, a fresh node-address uniform realizes the regular conditional source transition given that singleton $X_{\bar{h}}$ is not selected. This conditioning depends on the node state but not on an evaluation point. Both children are then generated recursively. An independent $X \sim P_X$ is drawn only after the full tree is constructed and is routed through it. Every actual or potential node is consequently an axis-aligned rectangle on a fixed root sample. One relative-VC event can cover all of these rectangles without conditioning on a selected target path. Indexing iid source draws by node address is a distributional construction and makes no claim about the package's sequential pseudorandom-number stream under a numerical seed.

Fix a node C , condition on $Q_M = q$, and rank the classes as in Section 3.3. Class $g_j(C)$ wins exactly when the candidate set contains no column from classes $g_1(C), \dots, g_{j-1}(C)$ and at least one column from class $g_j(C)$. Hence

$$\mathbb{P}\{g_j(C) \text{ wins} \mid Q_M = q, C\} = \frac{\binom{M-M_{j-1,C}^{\text{cum}}}{q} - \binom{M-M_{j,C}^{\text{cum}}}{q}}{\binom{M}{q}},$$

because the two terms in the numerator count, respectively, the candidate sets that exclude the preceding classes and those that also exclude class $g_j(C)$. This proves (15). Conditioning on Q_M proves (16). The same count gives

$$\mathbb{P}\{g \text{ is absent} \mid Q_M = q\} = \frac{\binom{M-r_g}{q}}{\binom{M}{q}},$$

which proves (17).

In either representation $R_{h,n}$, the other coordinate $X_{\bar{h}}$ is a singleton. Hence

$$\mathbb{P}\{X_{\bar{h}} \text{ is included} \mid Q_{M_n} = q\} = 1 - \frac{\binom{M_n-1}{q}}{\binom{M_n}{q}} = \frac{q}{M_n}. \quad (40)$$

Taking expectations gives the singleton's per-node inclusion probability $\omega_n = \mathbb{E}Q_{M_n}/M_n$.

To relate the source and population split criteria, recall that version 2.4.0 implements scalar `causal_forest` fits through the `instrumental_trainer` path [GRF Development Team, 2024]. The source class is `InstrumentalRelabelingStrategy`. Under (8), define

$$\Gamma = 4(W - 1/2)\{Y - m_a(X)\}, \quad \widehat{\Gamma} = 4(W - 1/2)\{Y - \widehat{m}(X)\}.$$

Random assignment and $\mathbb{E}(U \mid X, W) = 0$ give

$$\mathbb{E}(\Gamma \mid X = x) = \tau_a(x), \tag{41}$$

and uniform nuisance consistency gives

$$\|\widehat{\Gamma} - \Gamma\|_\infty \leq 2\|\widehat{m} - m_a\|_\infty = o_p(1). \tag{42}$$

For a fresh observation, conditioning on the nuisance-training state therefore yields

$$\left\| \mathbb{E}(\widehat{\Gamma} \mid X = \cdot, \mathcal{D}_n^+) - \tau_a \right\|_\infty = o_p(1). \tag{43}$$

Set $\ell_n = \log(B_n s_n)$ and

$$\epsilon_n^{\text{split}} = \sqrt{\ell_n/k_n} + \ell_n/k_n + \|\widehat{m} - m_a\|_\infty, \quad \epsilon_n^{\text{leaf}} = \sqrt{\ell_n/k_n}.$$

Both rates vanish under Assumption 3. Axis-aligned rectangles, candidate children, and their intersections with treatment arms form a fixed finite union of VC classes. Relative VC Bernstein bounds, followed by a union bound over the B_n splitting and estimation samples, apply simultaneously to every actual or potential node. Raw column multiplicity does not enlarge these geometric classes. Since every admissible child contains at least k_n observations from each treatment arm, the same event gives, uniformly over all admissible children A ,

$$\left| \frac{\mathbb{P}_n^{\text{split}}(\widehat{\Gamma} \mathbf{1}_A)}{\mathbb{P}_n^{\text{split}}(A)} - \mathbb{E}(\Gamma \mid A) \right| = O_p(\epsilon_n^{\text{split}}).$$

Consequently, the CART criterion based on $\widehat{\Gamma}$ converges uniformly to its population counterpart $\mathcal{Q}_C(j, t)$.

It remains to compare this criterion with the source criterion. Direct expansion of (11) gives

$$4\Psi_{i,C} = \widehat{\Gamma}_i - \widehat{\theta}_C + E_{i,C},$$

where

$$E_{i,C} = -4Z_i\tilde{\tilde{Y}}_C - 4\bar{Z}_C\tilde{Y}_i + 4\bar{Z}_C\tilde{\tilde{Y}}_C + 8\hat{\theta}_C Z_i\bar{Z}_C - 4\hat{\theta}_C\bar{Z}_C^2.$$

The same VC event, applied to Z , $Y - m_a(X)$, and the nuisance error, implies

$$\sup_C \{|\bar{Z}_C| + |\tilde{\tilde{Y}}_C|\} = O_p(\epsilon_n^{\text{split}}), \quad \sup_C |\hat{\theta}_C| = O_p(1).$$

Thus $\sup_C \max_{i \in C} |E_{i,C}| = O_p(\epsilon_n^{\text{split}})$. The term $-\hat{\theta}_C$ is constant within the parent and cancels from every child-mean contrast. Expanding the two bounded split criteria now gives

$$\sup_{C,j,t} |\hat{\mathcal{Q}}_C^{\text{grf}}(j,t) - \mathcal{Q}_C(j,t)| = O_p(\epsilon_n^{\text{split}}) = o_p(1). \quad (44)$$

Consider coordinate j before its first selection on a path. Its projection in the current rectangular node is $[0, 1]$. Conditional independence makes all other effect components cancel from a split contrast. For a split at t , the population criterion is

$$\mathcal{Q}_C(j,t) = \begin{cases} a_j^2 \frac{t}{1-t}, & 0 < t \leq 1/2, \\ a_j^2 \frac{1-t}{t}, & 1/2 \leq t < 1. \end{cases} \quad (45)$$

It has unique maximizer $1/2$ and maximum a_j^2 . The gain gap between the two unresolved coordinates is at least $\Delta_a(2\underline{a} + \Delta_a) > 0$. Set

$$\delta_* = \frac{1}{8} \min\{\underline{a}^2, \Delta_a(2\underline{a} + \Delta_a)\}.$$

By (44), the uniform criterion error is below δ_* with probability tending to one. This event preserves both the unresolved-coordinate ordering and the gap between an untouched signal and the residual gain of a previously selected signal.

Fix $\eta > 0$. At a node with $n_C \geq (4 + \eta)k_n$, the relative VC bounds make a sample threshold $1/2 + o_p(1)$ source admissible uniformly. If $4k_n \leq n_C < (4 + \eta)k_n$, every admissible child has at least $2k_n$ observations. Both empirical child fractions therefore lie in $[2/(4 + \eta), 1 - 2/(4 + \eta)]$, and uniform convergence of the untouched coordinate's conditional empirical distribution forces every admissible threshold to be within $O(\eta) + o_p(1)$ of $1/2$. The positive argmax gap in (45), (44), and the order of limits $n \rightarrow \infty$ followed by $\eta \downarrow 0$ give

$$\max_{\mathcal{T},j,F} |\hat{t}_{j,F} - 1/2| = o_p(1), \quad (46)$$

where $j \in \{1, 2\}$, $\hat{t}_{j,F}$ is the threshold selected when coordinate j is first used at node F , and

$\mathcal{T}$ ranges over every actual tree and potential failure tree. The maximum is zero when the set is empty.

To control later splits after one coordinate has been selected, let $b_j(C) = \min\{\mathbb{P}(\iota_j = 0 \mid C), \mathbb{P}(\iota_j = 1 \mid C)\}$ for a rectangular node C . Conditional independence and total variance imply

$$\sup_t \mathcal{Q}_C(j, t) \leq 4a_j^2 b_j(C), \quad \sum_{\ell=1}^K P_X(C_\ell) b_j(C_\ell) \leq P_X(C) b_j(C)$$

for every rectangular partition $C_1, \dots, C_K$ of C . The second inequality follows by summing the smaller of the unconditional masses for $\iota_j = 0$ and $\iota_j = 1$ in each child.

Let $\mathcal{N}_{j,\mathcal{T}}$ be the disjoint nodes at which coordinate j is first selected on branches of tree $\mathcal{T}$, and set

$$\mathcal{E}_{j,\mathcal{T}} = \sum_{F \in \mathcal{N}_{j,\mathcal{T}}} P_X(F) |\widehat{t}_{j,F} - 1/2|. \quad (47)$$

Since $X_j \mid F \sim \text{Unif}[0, 1]$, the two children created at F have total probability-weighted b_j -impurity equal to $P_X(F) |\widehat{t}_{j,F} - 1/2|$. Thus every descendant frontier $\mathcal{P}$ satisfies

$$\sum_{C \in \mathcal{P}} P_X(C) \sup_t \mathcal{Q}_C(j, t) \leq 4\bar{a}^2 \mathcal{E}_{j,\mathcal{T}}. \quad (48)$$

The first-split nodes form an antichain. Hence $\mathcal{E}_{j,\mathcal{T}} \leq \max_F |\widehat{t}_{j,F} - 1/2| \leq 1/2$. Equation (46), bounded convergence, and iterated expectation give

$$\mathbb{E}_\Theta(\mathcal{E}_{j,\mathcal{T}} \mid \mathcal{D}_n^+) = o_p(1), \quad j = 1, 2. \quad (49)$$

The same conclusion holds for either potential failure tree because (46) is uniform over those trees.

Set $\gamma = \underline{a}^2/4$. After coordinate j has been selected, let $\mathcal{N}_{j,\mathcal{T}}^{\text{hit}}(\gamma)$ contain the first descendant node on each branch where another coordinate remains admissible and $\sup_t \mathcal{Q}_C(j, t) > \gamma$. These nodes form an antichain, so (48) gives

$$\sum_{C \in \mathcal{N}_{j,\mathcal{T}}^{\text{hit}}(\gamma)} P_X(C) \leq \frac{4\bar{a}^2}{\gamma} \mathcal{E}_{j,\mathcal{T}}. \quad (50)$$

An untouched signal has an admissible near-half split with gain at least $\underline{a}^2 + o(1)$. Outside these descendant subtrees, a previously selected signal has gain at most γ . The uniform bound (44) then makes the untouched signal the preferred admissible signal with probability tending to one.

Let $\mathcal{U}_n$ collect the uniform criterion, admissibility, and localization events above, so $\mathbb{P}(\mathcal{U}_n^c) = o(1)$. For representation h , let $\mathcal{B}_{h,n}^0(x)$ contain $\mathcal{U}_n^c$ and the event that the potential failure path of x enters one of the first-hitting subtrees. Equations (49) and (50) imply

$$\max_{h \in \{1,2\}} \mathbb{E}_{\Theta_h^0} [\mathbb{P}_X \{ \mathcal{B}_{h,n}^0(X) \mid \Theta_h^0, \mathcal{D}_n^+ \} \mid \mathcal{D}_n^+] = o_p(1). \quad (51)$$

Outside $\mathcal{B}_{h,n}^0(X)$, every admissible unselected signal included in the candidate set defeats each signal selected earlier on the path, except at the initial competing-component split and the fixed-width terminal boundary.

On paths where signal j is first selected, let $\hat{\iota}_{j,\mathcal{T}}$ be its empirical side label and let $\mathcal{R}_{j,\mathcal{T}}$ be the union of terminal cells below such splits. The disagreement mass at node F equals $P_X(F) |\hat{\iota}_{j,F} - 1/2|$, and therefore

$$\mathbb{E}_\Theta \mathbb{P}_X \{ X \in \mathcal{R}_{j,\mathcal{T}}, \hat{\iota}_{j,\mathcal{T}}(X) \neq \iota_j(X) \} = o_p(1).$$

For an independent X' , write $L_{\mathcal{T}}(x)$ for the terminal cell containing x . Inserting the leaf-constant empirical label and applying conditional Jensen gives

$$\begin{aligned} & \int_{\mathcal{R}_{j,\mathcal{T}}} |\mathbb{E} \{ \iota_j(X') \mid X' \in L_{\mathcal{T}}(x), \mathcal{T} \} - \iota_j(x)| dP_X(x) \\ & \leq 2\mathbb{P}_X \{ X \in \mathcal{R}_{j,\mathcal{T}}, \hat{\iota}_{j,\mathcal{T}}(X) \neq \iota_j(X) \}. \end{aligned} \quad (52)$$

Thus imperfect threshold localization contributes $o_p(1)$ in integrated loss.

We now control the number of split opportunities. Fix h and route x through the full failure tree $\mathcal{T}_{h,n}^0$. Its transitions use the original law over M_n columns conditional on failure to select $X_{\bar{h}}$. No column is deleted, and Q_{M_n} is never redrawn under a different dimension. The increasing recodings of X_h induce identical routing at corresponding thresholds. Let $n_d(x)$ be the number of splitting observations in the node containing x after d splits.

We establish the depth envelope on the uniform relative-VC event. The information restriction and uniform treatment balance imply that every admissible child obeys, for some fixed $\beta_0 \in (0, \beta)$,

$$\beta_0 n_d(x) - 1 \leq n_{d+1}(x) \leq (1 - \beta_0) n_d(x) + 1$$

simultaneously over all actual and failure-tree nodes. To justify continuation, consider any rectangular node with $n_d(x) \geq C_{\text{cont}} k_n$, where C_{cont} is a sufficiently large fixed constant. Its treatment share is $1/2 + o(1)$, uniformly over roots and rectangles. Continuity of the covariate distribution makes the observations in each coordinate distinct almost surely, so every sampled coordinate is nonconstant whenever the node contains at least two observations.

Its empirical median creates two children of size $n_d(x)/2 + O(1)$, and each child contains $n_d(x)/4 + o\{n_d(x)\}$ observations from each treatment arm. For a sufficiently large fixed C_{cont} , these counts exceed k_n . Moreover,

$$\frac{n_A(1/4 - \bar{Z}_A^2)}{n_d(x)(1/4 - \bar{Z}_C^2)} = \frac{1}{2} + o(1) > \beta$$

for both median children. The median of every sampled column is therefore source admissible. Since $Q_{M_n} \geq 1$, the candidate set contains such a split. Condition (12) then forces a positive criterion value and continuation. Nodes with $n_d(x) \geq C_{\text{cont}}k_n$ therefore cannot be terminal. No split is admissible when $n_d(x) < 4k_n$, because each child must contain at least k_n observations on both sides of the parent treatment mean. The uncertain range $4k_n \leq n_d(x) < C_{\text{cont}}k_n$ has fixed multiplicative width and contributes only a bounded number of additional splits under the displayed recursion. These statements hold on one event over every actual and full failure-tree node because the root samples are fixed before the trees are generated. Choosing

$$0 < c_\beta < |\log \beta_0|^{-1}, \quad C_\beta > |\log(1 - \beta_0)|^{-1}$$

uses the fastest permitted contraction β_0 for the lower depth bound and the slowest contraction $1 - \beta_0$ for the upper bound. These choices give, simultaneously for $h = 1, 2$,

$$\mathbb{P}\{c_\beta H_n \leq J_{h,n}^0(X) \leq C_\beta H_n\} \rightarrow 1. \quad (53)$$

The same lower and upper recursions apply to every actual path, regardless of which coordinate is selected.

For fixed constants $0 < c_{\text{leaf}} < C_{\text{leaf}} < \infty$, every reached terminal cell L has population mass

$$c_{\text{leaf}} \frac{k_n}{s_n} \leq P_X(L) \leq C_{\text{leaf}} \frac{k_n}{s_n} \quad (54)$$

with probability tending to one. For the upper bound, a node above a sufficiently large multiple of k_n admits a median split, and (12) forces continuation. The lower bound follows from the treatment-arm and information restrictions on the split that created the leaf. Consequently there are at most $c_{\text{leaf}}^{-1} s_n / k_n$ leaves per tree.

For the estimation half, apply the relative VC inequalities to every axis-aligned rectangle. Write

$$N_{b,n}^{\text{est}}(L) = \sum_{i \in I_{b,n}^{\text{est}}} \mathbf{1}\{X_i \in L\}.$$

For $L = L_{b,n}^R(x)$, this count equals $N_{b,n}^{R,\text{est}}(x)$ from Section 2. Together with (54), these inequalities imply, for fixed constants $0 < c_{\text{est}} < C_{\text{est}} < \infty$,

$$c_{\text{est}} k_n \leq N_{b,n}^{\text{est}}(L) \leq C_{\text{est}} k_n$$

for every nonempty honest leaf, with probability tending to one. Honesty pruning is therefore asymptotically inactive. Write $\hat{\Gamma}_i = \Gamma_i + \delta_{i,n}$, where $\max_i |\delta_{i,n}| \leq 2\|\hat{m} - m_a\|_\infty$ by (42). Conditional on the splitting tree, the oracle scores Γ_i in its independent estimation half are independent across observations and satisfy $\mathbb{E}(\Gamma_i | X_i) = \tau_a(X_i)$. Applying the relative-VC bound to these oracle scores and then adding the uniform nuisance remainder gives

$$\max_{b,L} \left| \frac{1}{N_{b,n}^{\text{est}}(L)} \sum_{i \in L \cap I_{b,n}^{\text{est}}} \hat{\Gamma}_i - \mathbb{E}\{\tau_a(X) | X \in L\} \right| = O_p \left(\sqrt{\frac{\ell_n}{k_n}} + \|\hat{m} - m_a\|_\infty \right) = o_p(1). \quad (55)$$

For such a leaf, write

$$\bar{Z}_{b,n}(L) = \frac{1}{N_{b,n}^{\text{est}}(L)} \sum_{i \in L \cap I_{b,n}^{\text{est}}} Z_i.$$

The same argument for Z gives

$$\max_{b,L} |\bar{Z}_{b,n}(L)| = O_p(\epsilon_n^{\text{leaf}}) = o_p(1). \quad (56)$$

The forest-weighted $\bar{Z}_\alpha(x)$ is an average of these leaf means. Boundedness of $\tilde{Y}$, (13), and (56) therefore yield

$$\max_{h \in \{1,2\}} \|\hat{T}_{h,n} - \check{T}_{h,n}\|_\infty = O_p(\epsilon_n^{\text{leaf}}) = o_p(1). \quad (57)$$

It remains to derive the target for each representation. Work under the full failure-tree law used to define $\rho_{h,n,t}^0$. Condition on the training sample and the history through the current potential state, and write $\bar{h} = 3 - h$. The singleton is available with probability ω_n by (40), independently of the past. It may enter the candidate set without being chosen for the split. Let $K_{h,n}^{\text{str}}(x)$ count internal split nodes at which an available singleton can fail because a higher-gain unresolved component wins or the node lies in the final fixed-width leaf-size boundary.

For $h = 1$, the singleton has the larger unresolved gain and therefore wins whenever it is available, outside the exceptional set. For $h = 2$, the repeated component has the larger unresolved gain. A split at which that component blocks the singleton also resolves the repeated component. Once that split has been taken, an available unresolved singleton defeats every previously selected component on $\mathcal{U}_n$ outside the exceptional subtrees identified

in the adaptive-resolution argument. The failure-tree depth recursion leaves at most a fixed number C_{bd} of internal nodes in the terminal boundary. Hence $K_{h,n}^{\text{str}}(X) \leq 1 + C_{\text{bd}}$ on $\mathcal{U}_n$ for both representations. Since $\mathcal{U}_n^c \subseteq \mathcal{B}_{h,n}^0(X)$ has probability $o(1)$, the random-depth envelope gives

$$\max_{h \in \{1,2\}} \frac{K_{h,n}^{\text{str}}(X)}{J_{h,n}^0(X)} \longrightarrow_p 0 \quad (58)$$

in integrated probability, with the ratio defined as zero when $J_{h,n}^0(X) = 0$.

For $t \leq J_{h,n}^0(X)$, the predictable difference $\omega_n - \rho_{h,n,t}^0(X)$ is nonnegative. At a structural split node, this difference is at most ω_n . At every other internal split node on the complement of $\mathcal{B}_{h,n}^0(X)$, the adaptive-resolution argument shows statewise that singleton availability is equivalent to successful singleton resolution, so the difference is zero. Therefore, under the same potential failure-transition law,

$$0 \leq J_{h,n}^0(X)\omega_n - \Lambda_{h,n}^0(X) \leq K_{h,n}^{\text{str}}(X)\omega_n + J_{h,n}^0(X)\omega_n \mathbf{1}\{\mathcal{B}_{h,n}^0(X)\}.$$

Every term in this pathwise bound is evaluated under the same potential failure-transition law, so no probability from the ordinary path law is substituted. Dividing by the normalization $1 + J_{h,n}^0(X)\omega_n$ gives, with the ratio interpreted as zero when $J_{h,n}^0(X) = 0$,

$$\begin{aligned} & \frac{J_{h,n}^0(X)\omega_n - \Lambda_{h,n}^0(X)}{1 + J_{h,n}^0(X)\omega_n} \\ & \leq \frac{K_{h,n}^{\text{str}}(X)}{J_{h,n}^0(X)} \frac{J_{h,n}^0(X)\omega_n}{1 + J_{h,n}^0(X)\omega_n} + \mathbf{1}\{\mathcal{B}_{h,n}^0(X)\} \leq \frac{K_{h,n}^{\text{str}}(X)}{J_{h,n}^0(X)} + \mathbf{1}\{\mathcal{B}_{h,n}^0(X)\}. \end{aligned}$$

The first term is $o_p(1)$ by (58), and the second is $o_p(1)$ under the joint law of X and the potential failure-transition randomizers by (51). Consequently,

$$\max_{h \in \{1,2\}} \frac{|\Lambda_{h,n}^0(X) - J_{h,n}^0(X)\omega_n|}{1 + J_{h,n}^0(X)\omega_n} \longrightarrow_p 0, \quad \max_{h \in \{1,2\}} \max_{1 \leq t \leq J_{h,n}^0(X)} \rho_{h,n,t}^0(X) \leq \omega_n, \quad (59)$$

where the inner maximum is zero when $J_{h,n}^0(X) = 0$. Iterated conditioning on the potential failure states gives the exact failure-path identity in (1).

The depth and failure-path statements in this proof use the capped-Poisson source law (9) with finite ξ_{M_n} . A candidate law that samples all M_n columns almost surely requires a different probability calculation. It remains to verify, including when M_n is fixed, that the repeated component is resolved before the actual path reaches its leaf. Its conditional

candidate-inclusion probability is

$$p_n^{\text{rep}} = 1 - \frac{\nu_{M_n}(1)}{M_n} \geq \frac{1}{2}.$$

The actual-path versions of the antichain and depth arguments show, with conditional probability $1 - o_p(1)$, that the path avoids the exceptional subtrees and has at least $\lfloor c_\beta H_n \rfloor$ internal nodes. Set $L_n = \lfloor c_\beta H_n \rfloor - C_{\text{bd}}$. On this event, the first inclusion of the repeated class either resolves it or loses to and resolves the higher-gain singleton. Its next inclusion resolves the repeated component. Successive conditioning on the fresh node-address draws therefore bounds the failure probability by

$$\mathbb{P}\{\text{Bin}(L_n, p_n^{\text{rep}}) \leq 1\} \leq (1 + 2L_n)2^{-L_n} = o(1).$$

Let $\mathcal{B}_{h,n}^{\text{cop}}(X)$ denote failure to resolve the repeated component before the leaf. The preceding bounds give

$$\max_{h \in \{1,2\}} \mathbb{E}_\Theta [\mathbb{P}_X \{\mathcal{B}_{h,n}^{\text{cop}}(X) \mid \Theta, \mathcal{D}_n^+\} \mid \mathcal{D}_n^+] = o_p(1), \quad (60)$$

including fixed M_n .

For a generic tree $\mathcal{T}$, write

$$m_{j,\mathcal{T}}(x) = \mathbb{E}\{s_j(X') \mid X' \in L_{\mathcal{T}}(x), \mathcal{T}\}.$$

If the singleton is unresolved, the terminal cell is a function of X_h outside the integrated exceptional event. Independence and centering then give $m_{\bar{h},\mathcal{T}}(x) = 0$. If a component is resolved, (52) controls the difference between its leaf mean and its sign at x . Together with (60), these facts give the componentwise bounds

$$\begin{aligned} \mathbb{E}_\Theta [\|m_{h,\mathcal{T}} - s_h\|_{L^1(P_X)} \mid \mathcal{D}_n^+] &= o_p(1), \\ \mathbb{E}_\Theta [\|m_{\bar{h},\mathcal{T}} - D_{\bar{h},n}^{R_{h,n}} s_{\bar{h}}\|_{L^1(P_X)} \mid \mathcal{D}_n^+] &= o_p(1). \end{aligned} \quad (61)$$

The losses are bounded, so the joint fresh-tree bounds imply the displayed conditional statements by the tower property and Markov's inequality. Combining (61) with (55) gives

$$\mathbb{E}_\Theta \left[\left\| \tilde{T}_n^{R_{h,n}} - D_{\bar{h},n}^{R_{h,n}} \tau_a - \{1 - D_{\bar{h},n}^{R_{h,n}}\} a_h s_h \right\|_{L^1(P_X)} \mid \mathcal{D}_n^+ \right] = o_p(1). \quad (62)$$

Equations (41)–(43) verify Assumption 1 with $\mu = 0$, $f_1 = a_1 s_1$, and $f_2 = a_2 s_2$. Equations (55), (61), and (62) verify the honest-score and component-reduction parts of Assump-

tion 2. The bounded independent tree groups in Assumption 3, together with (57), verify its reported-forest linearization.

The repeated component is already handled directly by (60). No separate failure kernel is needed for it. For the singleton, the survival-product argument in the proof of Theorem 1 applies to $\Lambda_{h,n}^0$. Combining its conditional resolution probability with the mixture (62), forest aggregation, and (57) gives the source-forest target.

If $\mathcal{I}_n \rightarrow 0$, the upper depth envelope (53) and (59) give $\Lambda_{h,n}^0(X) \rightarrow_p 0$, so the singleton resolution probability vanishes and (19) follows. If $\mathcal{I}_n \rightarrow \infty$, the lower envelope gives $\Lambda_{h,n}^0(X) \rightarrow_p \infty$, so the singleton is resolved with probability tending to one. Together with (60), this proves (20). For a fixed $a \in \mathcal{C}_a$ in the finite-intensity regime, $\omega_n \rightarrow 0$, and (18) together with (59) gives

$$\Lambda_{h,n}^0(X) \xrightarrow[\mathbb{P}_a^0]{p} \kappa d_h(a), \quad \max_{1 \leq t \leq J_{h,n}^0(X)} \rho_{h,n,t}^0(X) \xrightarrow[\mathbb{P}_a^0]{p} 0.$$

The survival-product argument yields singleton failure probability $\zeta_h(a) = e^{-\kappa d_h(a)}$. Substitution into (62) gives the limits (21)–(22).

The same coupling covers the increasing recodings used in the theorem. At a node, the source represents a threshold split by a column value t and sends an observation left when its value is at most t . For every $\varphi_{g\ell n}$, the corresponding transformed threshold is $\varphi_{g\ell n}(t)$, and $\varphi_{g\ell n}(x) \leq \varphi_{g\ell n}(t)$ exactly when $x \leq t$. The candidate partitions, admissibility checks, split gains, and routing are therefore identical within each recoding block. The proof above depends on a block only through its multiplicity, so all three limits remain unchanged. $\square$

Proof of Corollary 1. In the selection regime, for $g \in \{1, 2\}$, Markov's inequality and the margin $a_g \geq \underline{a}$ give

$$\mathbb{P}_X\{\text{sgn}(\widehat{T}_{g,n}(X)) \neq s_g(X)\} \leq \frac{2}{\underline{a}} \|\widehat{T}_{g,n} - a_g s_g\|_{L^1(P_X)} = o_p(1). \quad (63)$$

Since s_1, s_2 are independent Rademacher variables, $\mathbb{P}(s_1 s_2 < 0) = 1/2$. On this event, $|\tau_a(X)| = a_2 - a_1 \geq \Delta_a$. The triangle inequality and another use of Markov's inequality show that, except on a set of P_X -mass $o_p(1)$,

$$\min\{|\widehat{T}_{1,n}(X)|, |\widehat{T}_{2,n}(X)|\} \geq \underline{a}/2.$$

This proves (24) and (25).

Because $a_2 > a_1 > 0$, $\text{sgn}\{\tau_a(X)\} = s_2(X)$. Equation (63) therefore proves (26). For the

other representation,

$$\text{Reg}_a(\mathbf{1}\{s_1 = 1\}) = \mathbb{E}[\tau_a\{\mathbf{1}\{s_2 = 1\} - \mathbf{1}\{s_1 = 1\}\}] = \frac{a_2 - a_1}{2}.$$

The bounded outcome and policy-classification convergence transfer this value to $\hat{\pi}_{1,n}$, proving (27). $\square$

Proof of Corollary 2. For every M , the capped-Poisson rule satisfies the exact identity

$$\mathbb{E}Q_M = \xi_M + e^{-\xi_M} - \mathbb{E}(N_M - M)_+.$$

If $M \rightarrow \infty$ and $\xi_M \equiv \xi$, the last expectation converges to zero, which proves (32). If $\xi_M \rightarrow \infty$ and $\xi_M/M \rightarrow 0$, Poisson concentration makes the upper cap irrelevant and $\mathbb{E}Q_M/\xi_M \rightarrow 1$. Under (33), $\xi_M/\sqrt{M} \rightarrow 1$, which proves (34). A fixed finite representation has $\omega_n > 0$ constant and $H_n\omega_n \rightarrow \infty$, so it is in the recovery regime. $\square$

Proof of Proposition 1. Because G is fixed, axis-aligned rectangles and coordinate halfspaces in $[0, 1]^G$ have fixed VC dimension. The criterion-transfer, localization, first-hitting, honest-leaf, and source-prediction arguments in the proof of Theorem 2 therefore hold uniformly over $a \in \mathcal{C}_G$, all coordinates, and all G representations. The split shape remains (45), and (28) gives a uniform ordering gap among unresolved coordinates. Outside an integrated event of probability $o(1)$, an included unresolved coordinate with the largest gain is selected, its first threshold is $1/2 + o_p(1)$, and its leaf contamination is $o_p(1)$.

Fix $R_{g,n}^{(G)}$. Each singleton has conditional inclusion probability $\omega_{G,n} = \mathbb{E}Q_{M_{G,n}}/M_{G,n}$, while the repeated class has probability

$$p_{g,n}^{\text{rep}} = 1 - \mathbb{E}\left\{\frac{\binom{G-1}{Q_{M_{G,n}}}}{\binom{M_{G,n}}{Q_{M_{G,n}}}}\right\} \geq \frac{c_n}{c_n + G - 1} \geq \frac{1}{G}.$$

In the selection regime, the path has at most $C_\beta H_n$ internal nodes with probability tending to one. The probability that some singleton enters a candidate set on the path is at most $(G-1)C_\beta H_n \omega_{G,n} + o(1) = o(1)$. On its complement, all splits use the repeated class. Independence, centering, and the uniform leaf and prediction transfers give $\|\hat{T}_{g,n}^{(G)} - a_g s_g\|_{L^1(P_X)} = o_p(1)$. Maximizing over fixed G proves (29).

For recovery, the path has at least $c_\beta H_n$ internal nodes with probability tending to one. Divide them into G consecutive blocks of length $L_n \asymp H_n/G$. Fresh node-address draws and the preceding bound imply, for every split-action class j , that its probability of no inclusion in a given block is at most $\exp(-L_n \omega_{G,n}) = o(1)$. Order coordinates by decreasing

a_j^2 . Successive blocks resolve the largest-gain remaining coordinate outside the common exceptional event. A union bound over fixed G shows that all coordinates are resolved before stopping, which proves (30).

For $g \neq h$, (29) and the margin $a_j \geq \underline{a}_G$ imply $\mathbb{P}_X\{\text{sgn}(\widehat{T}_{j,n}^{(G)}) \neq s_j\} = o_p(1)$ for $j \in \{g, h\}$. Independence gives $\mathbb{P}(s_g s_h < 0) = 1/2$, proving (31). The multiplicity boundaries follow from the calculation for Corollary 2 after replacing M_n by $M_{G,n} = c_n + G - 1$. $\square$

Proof of Proposition 2. Couple two quotient-preserving representations by all the common randomization listed before the proposition. At any common node state, they draw the same classes, evaluate the same certified actions, apply the same scores and tie order, and select the same action. The weak-order certificate gives identical unordered children, and canonical orientation routes every point in $\mathcal{X}_{\text{cert}}$ to the same child. Induction gives identical tree partitions on the certified array. Leaf observations and forest weights then coincide. The measurability condition extends this equality to nuisance fits, stopping decisions, sufficient statistics, predictions, reported variance calculations, rankings, and policy ties. This proves the pathwise statement.

At node C , class g wins when it is sampled and none of the $h_g(C)$ classes preceding it in the total score order is sampled. Conditional on $\widetilde{Q}_K = q$,

$$\mathbb{P}\{g \text{ wins} \mid \widetilde{Q}_K = q, C\} = \frac{\binom{K-h_g(C)}{q} - \binom{K-h_g(C)-1}{q}}{\binom{K}{q}}.$$

The two terms in the numerator count candidate sets that exclude the preceding classes, with and without class g . Averaging over q proves (35). $\square$